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Usage of Gravity Assists and Aerobraking at Titan for a Mission to Enceladus

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Contents

List of Figures	4
List of Tables	6
1 Introduction	7
1.1 Motivation	7
1.2 Objectives	7
2 Astronomical Background	9
2.1 Orbital Dynamics	9
2.1.1 Basics of Interplanetary Travel	9
2.1.2 Gravity Assist and Aerobraking	12
2.1.3 Thermal Protection Systems	14
2.2 The Saturnian System	15
2.3 History of Aerobraking	18
3 Description of the Program Code	20
3.1 User Input	20
3.2 Preparing the Program	28
3.3 Atmospheric Model	29
3.4 Equations of Motion	30
3.5 Launch Delay Calculation	34
3.6 Adjust Aerobraking Altitude	36
4 Results	41
4.1 Simulation Results	41
4.2 Validation of Results	44
4.2.1 Validation of the Atmospheric Model	44
4.2.2 Validation of Gravitational Forces	47
4.2.3 Validation of Aerodynamic Forces	48
4.2.4 Validation of Aerothermodynamics	51
5 Conclusion	53
6 Bibliography	54
A Extra Figures	57
A.1 View of all Saturn Moons	57

A.2	Complete Trajectory Plot	58
A.3	Deviation in Aerothermodynamics	59
B	Reference Frames	63
B.1	Inertial Cartesian Reference Frame	63
B.2	Orbit Reference Frame	63
C	Python Code	64
C.1	Launch Delay Iteration Process	64
C.2	Adjust Aerobrake Altitude Process	64

List of Figures

2.1	Patched conics approximation (Parvathi et al., 2016)	10
2.2	Arrival phase visualization (Curtis, 2014)	11
2.3	Gravity assist energy increase (Brown, 1998)	12
2.4	Gravity assist energy decrease (Brown, 1998)	12
2.5	Saturn moon size comparison (Seal, 2002)	16
2.6	Saturn moon orbit (Winter, 2021)	17
2.7	Atmospheric density comparison (Justus et al., 2003)	18
3.1	Trajectory example (own work)	22
3.2	Atmospheric density vs. altitude (own work)	23
3.3	Data over time example (own work)	24
3.4	Possible altitudes for orbit capture (own work)	25
3.5	Possible aerobraking altitude for orbit capture (own work)	26
3.6	Heat loads during aerobraking (own work)	27
3.7	Atmospheric density model (own work)	30
3.8	Logic of the launch delay iteration (own work)	35
3.9	Delay iteration example (own work)	36
3.10	Database entry trajectory (own work)	37
3.11	Final print statement (own work)	38
3.12	Two vs. three flyby comparison (own work)	39
3.13	Logic of the aerobrake height iteration (own work)	40
4.1	Different flyby altitude trajectory comparison (own work)	42
4.2	Lowest possible aerobraking altitude (own work)	43
4.3	Atmospheric data from HASI (Jet Propulsion Laboratory, 2019)	45
4.4	Comparison of the atmospheric model and HASI data (own work)	46
4.5	Titan orbiter transfer trajectory (Nixon et al., 2016)	47
4.6	Titan orbiter arrival trajectory (Nixon et al., 2016)	47
4.7	Spacecraft velocity when reaching Titan (own work)	48
4.8	HYPAS aerocapture guide (Way et al., 2003)	49
4.9	Program aerocapture guide (own work)	50
4.10	Heatflux during aerocapture NASA study (Laub, 2012)	51
4.11	Heatflux during aerocapture program model	52
A.1	Saturn and all known moons (Barnett, 2025)	57
A.2	Complete Trajectory Plot (own work)	58
A.3	Atmosphere model NASA aerothermodynamics study (Laub, 2012)	59

A.4	Atmosphere model simulation program (own work)	60
A.5	Altitude & velocity during aerocapture NASA aerothermodynamics study (Laub, 2012))	61
A.6	Velocity during aerocapture simulation program (own work)	62
A.7	Altitude during aerocapture simulation program (own work)	62

List of Tables

2.1	Heat sink material properties (Haynes, 2017) & (own work)	14
3.1	Rocket parameters (own work)	20
3.2	Output parameters (own work)	21
3.3	Planetary and orbital constants (own work)	29
3.4	Spacecraft status parameters (own work)	31
3.5	Equations of motion formula symbols (own work)	33
3.6	Database entry (own work)	36

1 Introduction

1.1 Motivation

Enceladus is a small, icy moon in the Saturnian system that has gained considerable attention over the past few years due to numerous scientific discoveries. The most important being that Enceladus meets the three requirements for a habitable environment: the presence of liquid water, a source of energy, and a specific set of chemical elements (ESA, 2024). This makes Enceladus an optimal target for space exploration missions.

Yet, no space agency has managed to land on Enceladus to date. One of the reasons for that is the massive amount of deceleration required in the Saturnian system to establish orbit around it. Traditional deceleration methods relying on thrusters require large amounts of fuel, thus limiting the mass available for scientific payloads.

This thesis investigates an alternative approach. Gravity assists and aerobraking maneuvers have been used in previous missions at Venus and Mars and proven to be effective and reliable ways of changing a spacecraft's orbit without the need for fuel (ESA, 2017). Enceladus' small mass and non-existent atmosphere, however, are challenges that need to be overcome. Leveraging another celestial body's gravity and atmosphere for aerobraking, it might be possible to reduce fuel requirements and allow for better mission objectives in future missions towards Enceladus.

1.2 Objectives

This thesis will elaborate on a computer program used to simulate different trajectories around Saturn, including its moons Titan and Enceladus. The results of that program will be analyzed to assess the possibilities of establishing an orbit around Enceladus utilizing aerobraking and gravity assists and the effects of such maneuvers on the mission duration and orbiter structure due to generated heat loads. The problem to solve in this thesis is finding multiple trajectories for a variety of arrival velocities. The spacecraft's starting

point is located at the edge of Saturn's sphere of influence. The exact starting location depends on the arrival velocity, and is set so that the crossing point with Titan's orbit on its first flyby is identical, ensuring comparable results. The final orbit to be attained has its periapsis within the sphere of influence of Enceladus and the apoapsis close to Titan. The exact apoapsis height will vary depending on the velocity, since the orbital period should be synchronized with Enceladus, ensuring that each time the spacecraft reaches periapsis, Enceladus is also present for potential rendezvous or orbit insertion opportunities. Several variables are optimized, such as time of flight, thermal loads generated during aerobraking, and the resulting mass gain of the heat protection systems. Although a lower arrival velocity seems logical to reduce deceleration requirements, Saturn's significant gravitational influence gives a lower boundary on the slowest arrival velocity feasible. Arriving with a low velocity in the sphere of influence risks a trajectory that gets pulled into Saturn and ends in mission loss. Hence, a balance between reducing approach speed and maintaining a viable capture trajectory is required. According to this, the arrival velocities examined are between 0.5 km/s and 3.5 km/s.

2 Astronomical Background

2.1 Orbital Dynamics

2.1.1 Basics of Interplanetary Travel

Interplanetary travel consists of at least four bodies:

1. Departure body - in this case Earth
2. Arrival body - in this case Enceladus
3. The Sun
4. The spacecraft

Calculating the forces of all bodies onto each other is called an N-body simulation. Using it for the four bodies listed above is almost impossible due to technical limitations. Hence, mission planning typically employs the patched conics approximation. In this method, the mission is divided into the three distinct phases of departure, transfer and arrival. Each phase only consists of the spacecraft and the dominant gravitational source during that phase. Therefore the problem gets split into multiple two-body problems which are significantly easier to solve. The transition between these phases is determined by the concept of the sphere of influence (SOI). A sphere of influence is the region around a celestial body where its gravitational pull exceeds others and therefore is dominant. The radius of the sphere of influence is given by:

$$r_{SOI} = a \left(\frac{\mu_{body}}{\mu_{Sun}} \right)^{\frac{2}{5}} \quad (2.1)$$

where a is the semi-major axis of the celestial body's orbit, μ_{body} is the gravitational parameter of the body, for example Earth, and μ_{Sun} is the Sun's gravitational parameter.

Figure 2.1 illustrates the three-phase approximation applied to an interplanetary journey between Earth and Mars.

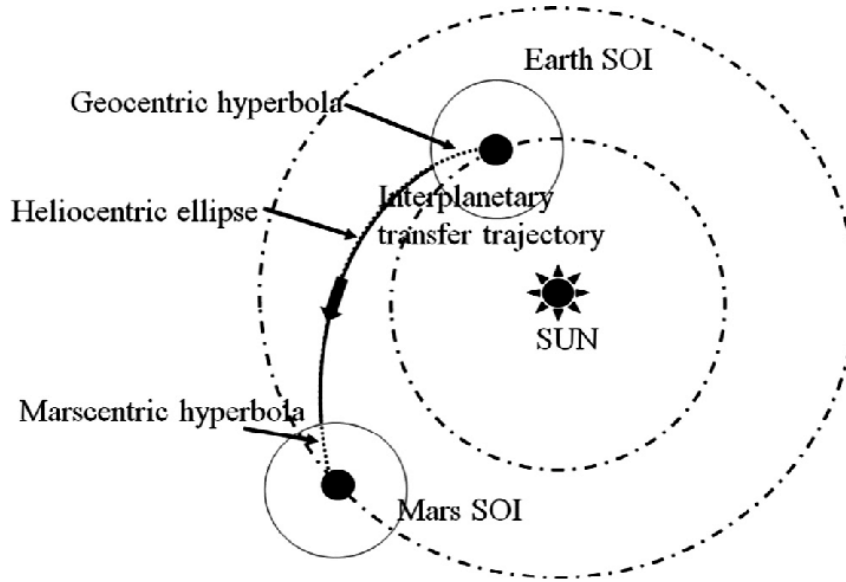


Figure 2.1: Patched conics approximation (Parvathi et al., 2016)

For the examined mission in this thesis there are two more bodies; Saturn and Titan. This makes an N-body simulation even less feasible. Using the patched conics approximation allows to divide the mission and focus only on the arrival phase and still receive valuable results and insights.

In interplanetary travel, the velocity of the spacecraft is one of the most important parameters to track and adjust. A spacecraft arriving in the Saturnian system has excessive arrival velocities, requiring deceleration to capture it into a stable orbit. Figure 2.2 shows the arrival of a spacecraft at Mars. The flyby trajectory marked in green is the path a spacecraft would take without any sort of deceleration using thrust or aerobraking, while the capture orbit marked in blue is the target trajectory of the spacecraft.

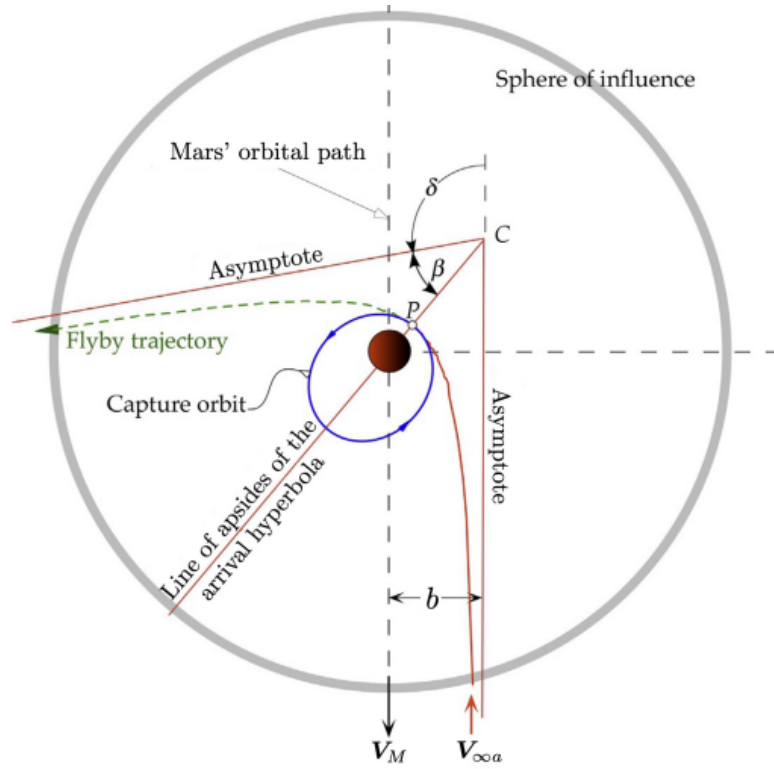


Figure 2.2: Arrival phase visualization (Curtis, 2014)

To go from the green flyby trajectory to the blue capture orbit, the spacecraft needs to be decelerated at the point P called periapsis. It is the point on the trajectory where the spacecraft is closest to the central body of an orbit (Curtis, 2014). The velocity is fundamentally defined by the vis-viva equation:

$$v = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}} \quad (2.2)$$

Since the gravitational parameter μ and the distance of the spacecraft to the center of mass r have not changed, the semi-major axis (a) must change. The semi-major axis is defined as:

$$a = \frac{r_p + r_a}{2} \quad (2.3)$$

Since the spacecraft is located at the periapsis r_p , the only parameter that can change is the farthest point of the orbit, the apoapsis r_a (Curtis, 2014).

2.1.2 Gravity Assist and Aerobraking

A gravity assist describes a maneuver of a spacecraft using the gravitational force of a celestial body for its benefit. It is mostly used to accelerate or decelerate a spacecraft on its trajectory and significantly alters the trajectory of the spacecraft after the maneuver. Flying ahead or in front of a celestial body on its orbit, the spacecraft experiences a gravitational force of said body that acts in the opposing direction of the spacecraft's velocity vector. This results in the spacecraft losing energy and reducing its velocity. The same principle works for flying behind a celestial body on its orbit, resulting in an increase of velocity. Figure 2.3 and Figure 2.4 show both cases.

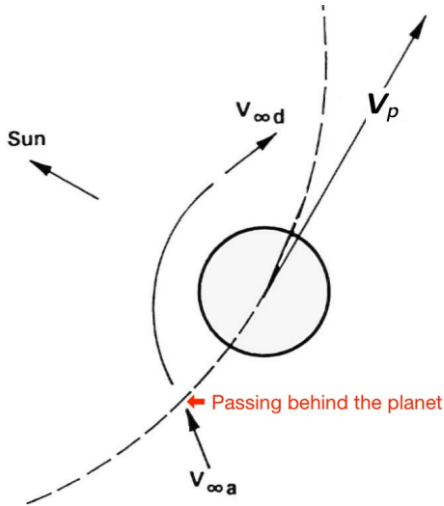


Figure 2.3: Gravity assist energy increase (Brown, 1998)

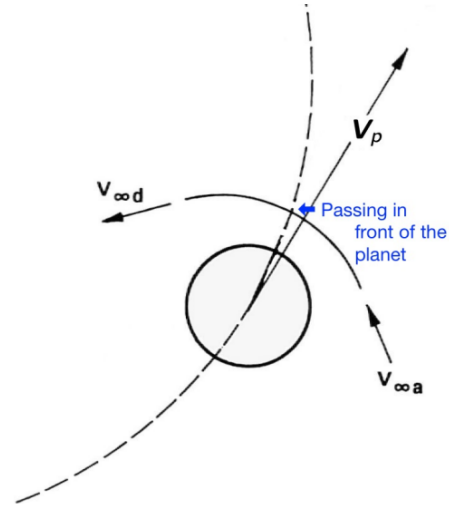


Figure 2.4: Gravity assist energy decrease (Brown, 1998)

The vector of force $\mathbf{F}$ acting onto the spacecraft can be described by Newton's law

of universal gravitation (Roy, 2005). Using M as the mass of the planet, m as the spacecraft's mass and $\hat{\mathbf{r}}$ as the unit vector giving the direction, the force is described by the following equation:

$$\mathbf{F} = -\mathcal{G} \frac{Mm}{r^2} (-\hat{\mathbf{r}}) \quad (2.4)$$

Since the gravitational force that one body exerts on another decreases with the square of the distance between them, the change in velocity is dependent on the distance of the flyby maneuver. It is imperative to note that Newton's laws only apply in an inertial frame. An inertial frame is neither rotated nor accelerated. More to reference frames can be found in Appendix B.

On a celestial body with an atmosphere, an even more advanced reduction in velocity is possible. Aerobraking describes a maneuver of a spacecraft using the atmospheric drag experienced when flying through an atmosphere at certain altitudes to reduce its velocity.

c_d is a unit less coefficient that describes how aerodynamic a body is. Together with the density of the atmosphere ρ , the cross-sectional area A and the velocity v , the force of the air drag can be calculated with:

$$D = \frac{1}{2} c_d \rho A v^2 \quad (2.5)$$

Because this force acts opposite to the direction of travel, it further decreases the spacecraft's kinetic energy. This maneuver is not feasible on all celestial bodies. For a celestial body with a very low atmospheric density the impact on trajectory may not be strong enough or the required altitude to the body is too low to keep safety margins. Since aerobraking maneuvers are significantly closer to the celestial body compared to gravity assists, errors can potentially result in a loss of mission. Hence, aerobraking requires significantly more precise trajectory design and planning. Aerobraking, when done correctly, however, can have a tremendous positive impact on the mission.

2.1.3 Thermal Protection Systems

There are three ways to protect a spacecraft from thermal loads:

1. Radiative thermal solution via coatings offers a constant dissipation of heat with no additional weight. It is the only thermal protection system where the time spent in the atmosphere does not matter. This makes it the optimal thermal solution if the heat loads allow for it. Modern coatings, designed for in-atmospheric use, are capable of keeping an emissivity of 0.9 up to 1773 K resulting in up to 504,266 W/m² (Shao et al., 2019).

$$q = 0.9 \times 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \text{ K}^4) \times (1773 \text{ K})^4 = 50.4267 \text{ W}/\text{cm}^2 \quad (2.6)$$

2. Heat sink solutions store the heat during atmospheric phases and dissipate it during the rest of the orbit. Aluminum and copper are common materials used for heat sinks. Aluminum, however, offers a better ratio of thermal conductivity when put into perspective of its density and therefore the weight of the heat sink. The material properties from ?? and Equation 2.7 deliver the following results for heat sinks:

Property	Aluminum	Copper	Unit	Formula symbol
Specific heat capacity	0.897	0.385	kJ/(kg K)	c
Material density	2700	8960	kg/m ³	ρ
Allowed temperature change	100	100	K	K
Material thickness	100	100	mm	t
Possible Heat load	2421.9	3434.2	J/cm	Q

Table 2.1: Heat sink material properties (Haynes, 2017) & (own work)

$$Q = c \rho t \Delta T \quad (2.7)$$

3. Ablative thermal protection systems heat up until they vaporize to dissipate the heat. The current state-of-the-art material used for this application is Phenolic Impregnated Carbon Ablator, also known as PICA, and offers a specific heat capacity $c_p(T)$ between 983.9 J/(kg K) at 298 K up to 2001 J/(kg K) at 2222 K and an enthalpy of pyrolysis $\Delta H_{pyrolysis}$ of 29.5 MJ/kg (Wang et al., 2019). The total heat required to vaporize PICA therefore results in 69.4 MJ/kg. Using a density ρ of 266 kg/m³, an ablative thermal protection system made from a 100 mm thick t PICA material can handle up to 1846 MJ/m².

$$Q_{total} = \left(\int_{T_0}^{T_{vaporization}} c_p(T) dT + \Delta H_{pyrolysis} \right) \rho t \quad (2.8)$$

The next generation of ablative thermal protection will be the Heatshield for Extreme Entry Environment Technology (HEEET) that is currently still in development. NASA plans on using HEEET for future missions to Saturn, so it is likely that it will find use in the first missions toward Enceladus and be able to withstand even higher heat loads (Gasch et al., 2022).

2.2 The Saturnian System

The Saturnian system consists of Saturn along with its currently known 146 moons. The last discovery of new moons in the Saturnian system was in 2020, making it likely that there are other moons yet to be discovered. This makes it the planet with most moons known to mankind. A full view of Saturn and all known moons can be found in the appendix at Figure A.1. The moons range from Aegaeon with a mean radius of 0.12 km up to Titan with an equatorial radius of 2575.5 km being larger than Mercury (Lindsey, 2024). The size of the moons is visualized in Figure 2.5

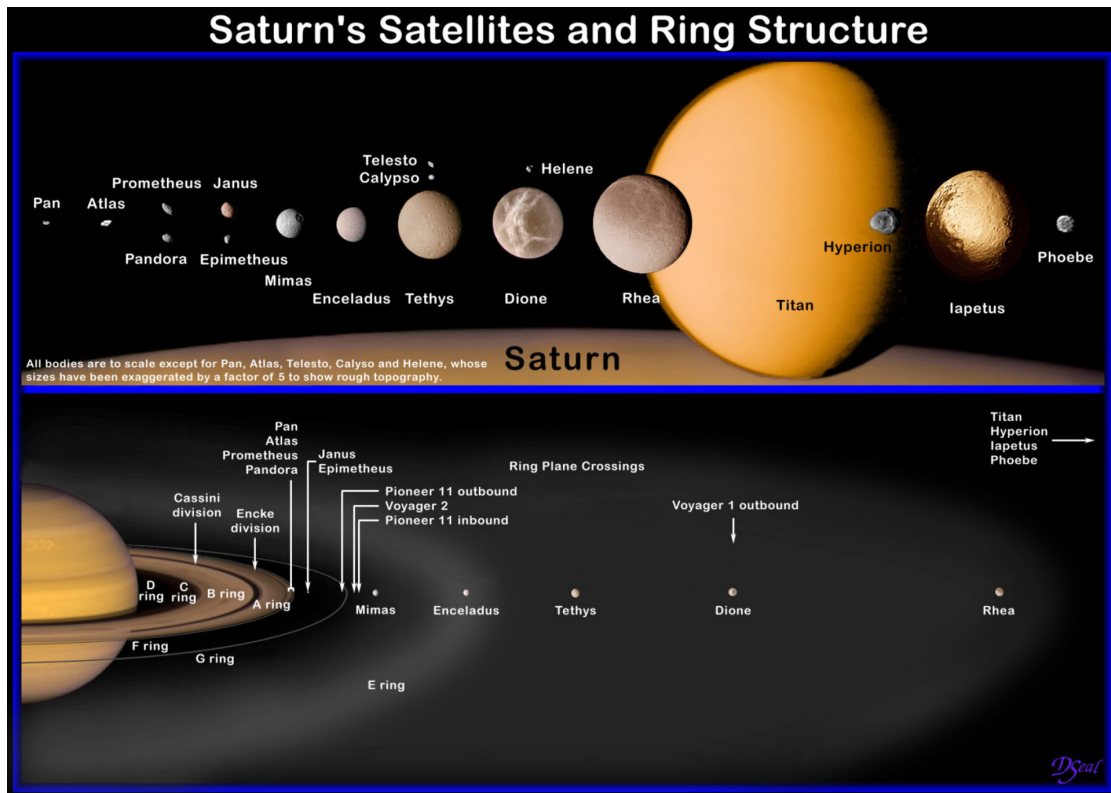


Figure 2.5: Saturn moon size comparison (Seal, 2002)

Of the 7 major moons, Titan is the second furthest away, only Iapetus being more than twice as distant to Saturn. Enceladus in comparison is the second closest to Saturn. Figure 2.6 delivers an overview of the orbits of Saturn's major moons.

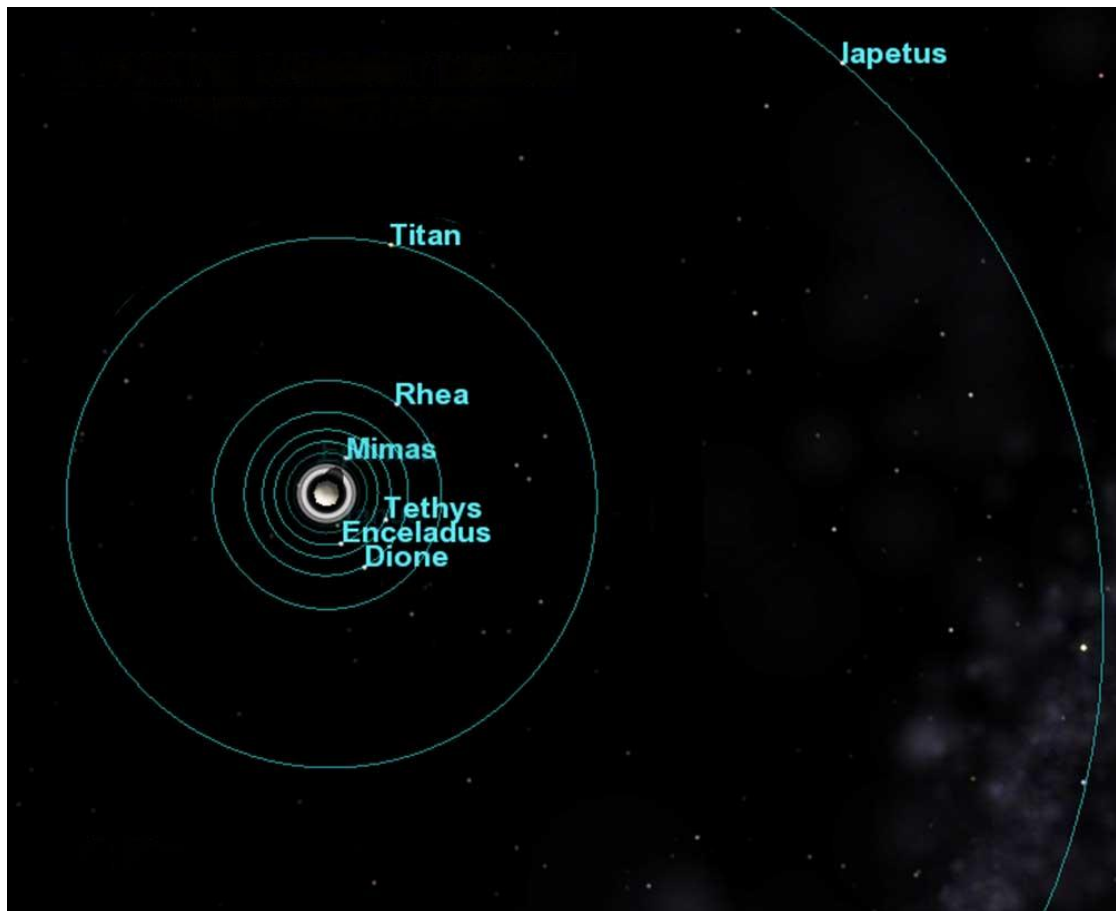


Figure 2.6: Saturn moon orbit (Winter, 2021)

Although the target planet of this thesis is Enceladus, Titan does have a significant role in reaching it. Enceladus' small mass results in a sphere of influence radius of about only 488 km, making both aerobraking as well as gravity assist not a viable option. The Saturnian system, however, does offer a different moon for aerobraking. Titan's atmosphere has been characterized by the Cassini-Huygens mission in 2005. The atmospheric density profile and mass of Titan theoretically allow for a different variety of aerobraking and gravity assist maneuvers around it (Jet Propulsion Laboratory, 2019). Not only is it the only known moon to have a substantial atmosphere, but its atmosphere is also a lot more dense at higher altitudes compared to other planets. Figure 2.7 shows that the required altitude above Titan for aerobraking is 700 km, while the same density is found at altitudes of 100 km or lower on Mars.

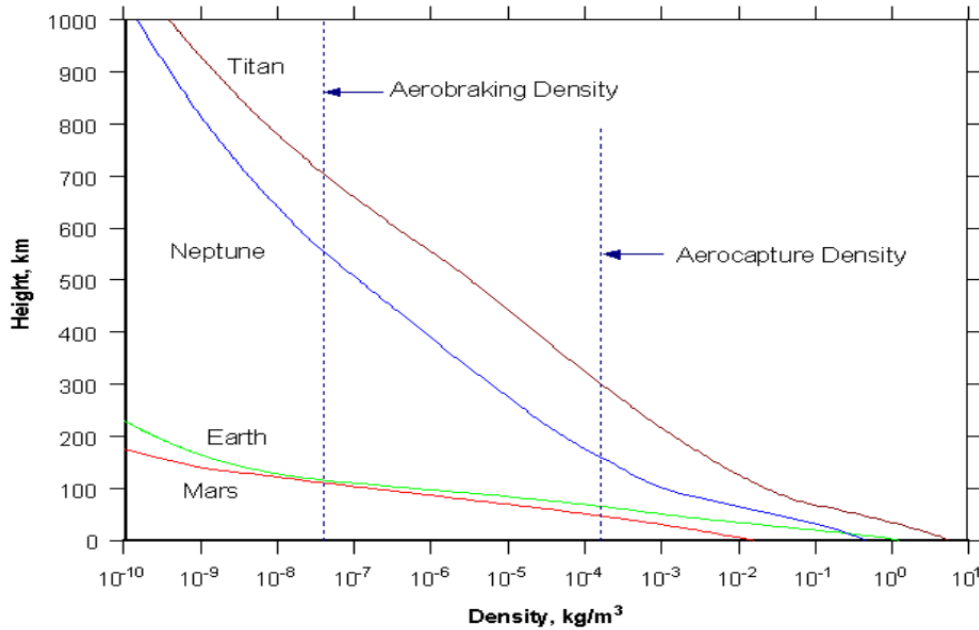


Figure 2.7: Atmospheric density comparison (Justus et al., 2003)

This gives aerobraking or aerocapture missions at Titan a larger safety margin for perturbations not predicted by a simulation and therefore decrease the risk of mission failure.

2.3 History of Aerobraking

In the past 30 years, aerobraking and gravity assists have become reliable ways of adjusting a spacecraft's trajectory without having to spend fuel. It has become a central strategy in modern mission planning to minimize fuel costs and maximize payload. The first tests to see if the idea of aerobraking was feasible happened in 1991. The Hiten spacecraft from ISAS from Japan used Earth's atmosphere to slowdown by 1.71 m/s (Ahrens, 2021). Since then it has been successfully used in various missions around Venus and Mars and has not only reduced the fuel requirements of those missions but has also enabled missions that were unattainable without the velocity reduction provided by the aerobraking maneuver like the Magellan mission on Venus in 1993 (Jet Propulsion Laboratory, 1993). In 2014 the European Space Agency also used aerobraking for the

first time with the Venus Express (ESA, 2014). After the successful test, the ESA then used aerobraking to lower the orbit of the ExoMars Trace Gas Orbiter mission from 33,000 km to 400 km, mainly using the atmosphere of Mars (ESA, 2017).

3 Description of the Program Code

3.1 User Input

The user input is the first region of the program and is used to modify the simulation to be as close to the planned mission as possible. It is divided into Rocket parameters and Output parameters.

The rocket parameters define the properties of the spacecraft. The parameters in Table 3.1 have a significant impact on the trajectory and therefore must be adjusted before use to ensure realistic simulation results.

Constant	Codename	Value	Unit
Total mass	m_0	3725.0	kg
Propellant mass	m_p	2725.0	kg
Thrust	thrust	0	N
Hyperbolic excess velocity	v_inf	0.5 - 3.5	km/s
Specific Impulse	Isp	3100	s
Drag Coefficient	c_d	2.2	—
Cross-Sectional Area	A	11.05	m ²
Heat Load Coefficient	c_q	0.5	—

Table 3.1: Rocket parameters (own work)

Total mass is the total mass of the spacecraft that enters Saturn's sphere of influence. Propellant mass is the amount of propellant before any thrust maneuvers. Thrust is the maximum thrust provided by the propulsion system of the spacecraft. Hyperbolic excess velocity is the spacecraft's velocity relative to Saturn upon entering its sphere of influence. Specific Impulse measures the amount of thrust produced per unit of fuel. Drag Coefficient is a dimensionless coefficient indicating how aerodynamically

optimized the spacecraft is. Cross-Sectional Area is the area of the spacecraft exposed to the atmosphere. The Heat load coefficient is a coefficient used to calculate the heat loads on the spacecraft during atmospheric flight phases. 0.5 is the maximum possible value; therefore, any real-world application data should not exceed this value.

The output parameters allow the user to display only those plots relevant for the mission analysis and to adjust the simulation duration. Table 3.2 shows the available settings in the program.

Option	Codename	Value	Unit
Simulation duration	tint	1500	d
Titan Orbit Crossings	max_crossings	8	—
Plot Trajectory	Plot_Trajectory	True/False	—
Plot Atmosphere	Plot_Atmosphere	True/False	—
Plot Data over time	Plot_Values	True/False	—
Plot Aerobrake Heights	Plot_Comparison	True/False	—
Plot Heat loads	Plot_Heating_Rate	True/False	—

Table 3.2: Output parameters (own work)

Simulation duration is the maximum time (in days) for which the simulation runs. Titan Orbit Crossings provide an alternative way of limiting simulation time based on the number of times the spacecraft crosses Titan’s orbit. Once the specified number of crossings is reached, the simulation stops. If Plot Trajectory is set to True, a plot of all simulated trajectory configurations is shown. Selecting a checkbox for a specific configuration will display its trajectory. A section of that plot is shown in Figure 3.1. The yellow circle represents Saturn, the light blue circle around Saturn represents Enceladus’ orbit, the light red circle represents Titan’s orbit, and the blue line represents the spacecraft’s trajectory. A slider below the plot allows to depict the position of the spacecraft, Titan, and Enceladus at each time. The complete figure, including Saturn’s sphere of influence, is located in the appendix at Figure A.7.

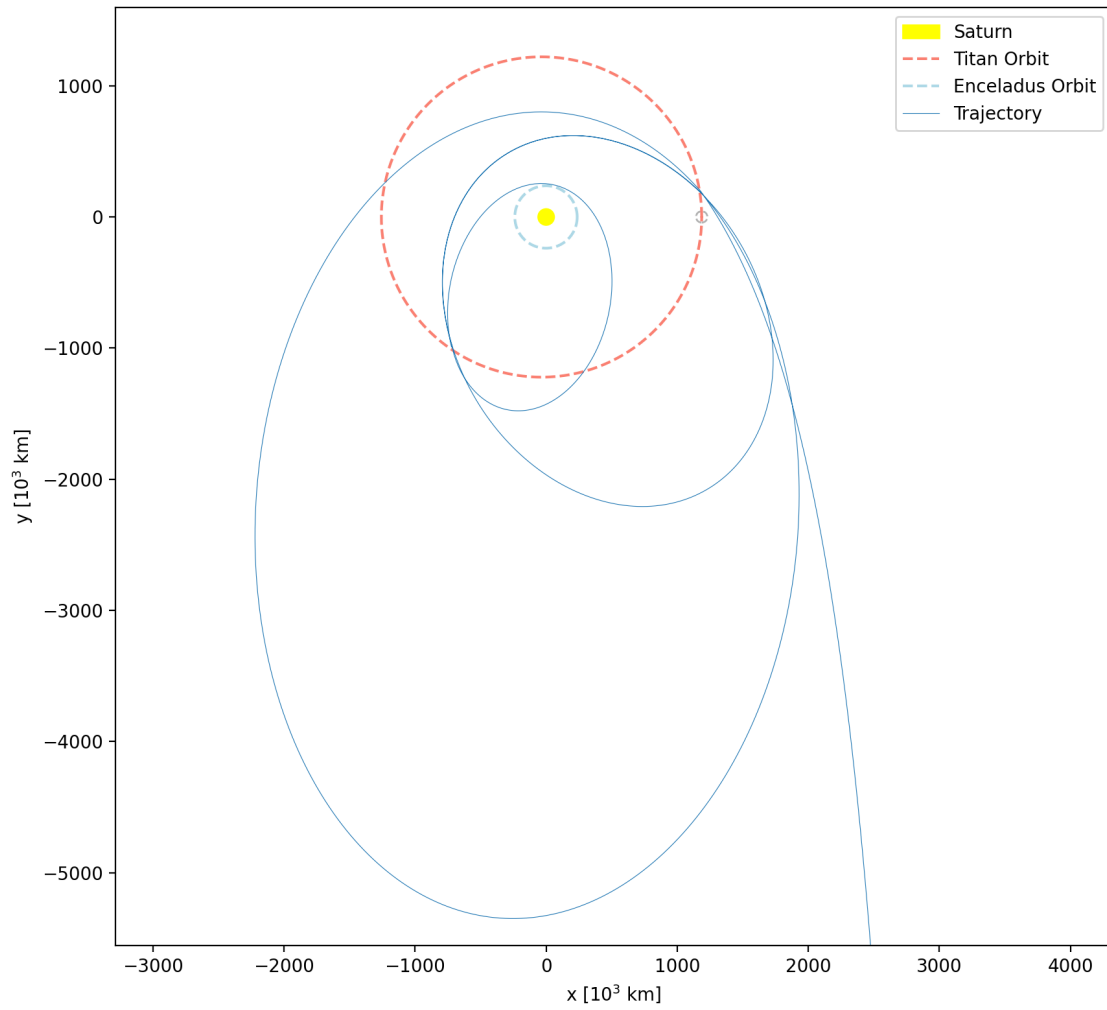


Figure 3.1: Trajectory example (own work)

The atmospheric density based on altitude, shown in Figure 3.2, is displayed when the Plot Atmosphere option is set to True.

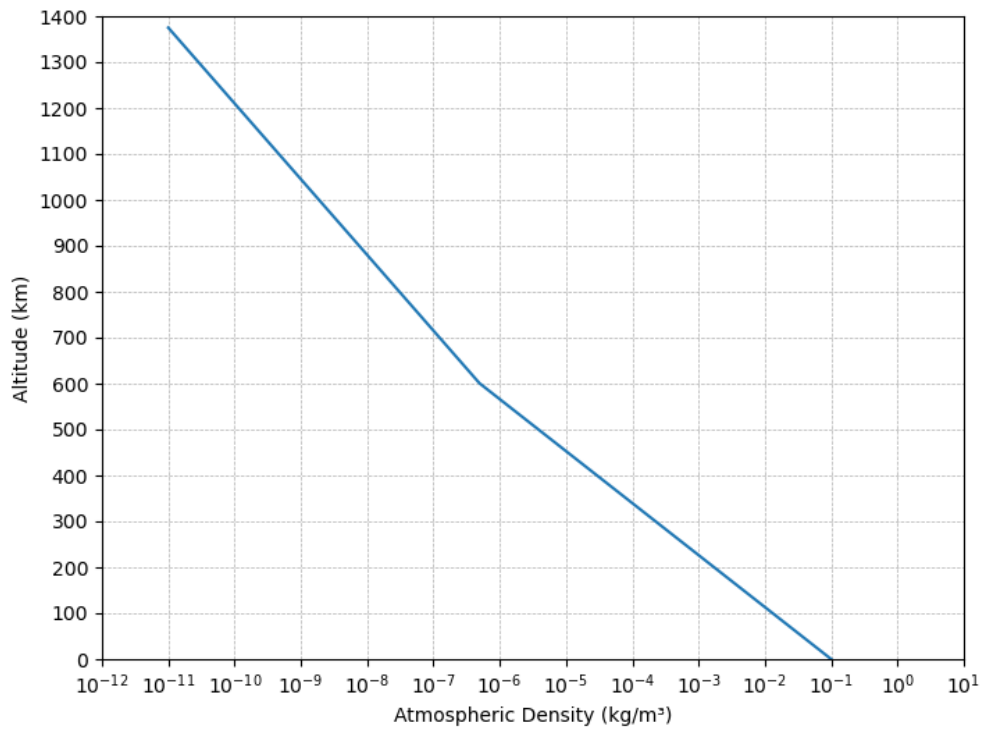


Figure 3.2: Atmospheric density vs. altitude (own work)

If Plot Data over time is enabled, the program displays various mission parameters over time. On the left side of the interface is a list of configurations, each selectable by a checkbox. On the right side is a list of tracked parameters, including orbital energy, distance to Saturn, distance to Titan, and velocity. Time periods during which the spacecraft is within Titan's atmosphere are shaded in light red, and times when thrust is active are highlighted in light blue. An example of such data is shown in Figure 3.3.

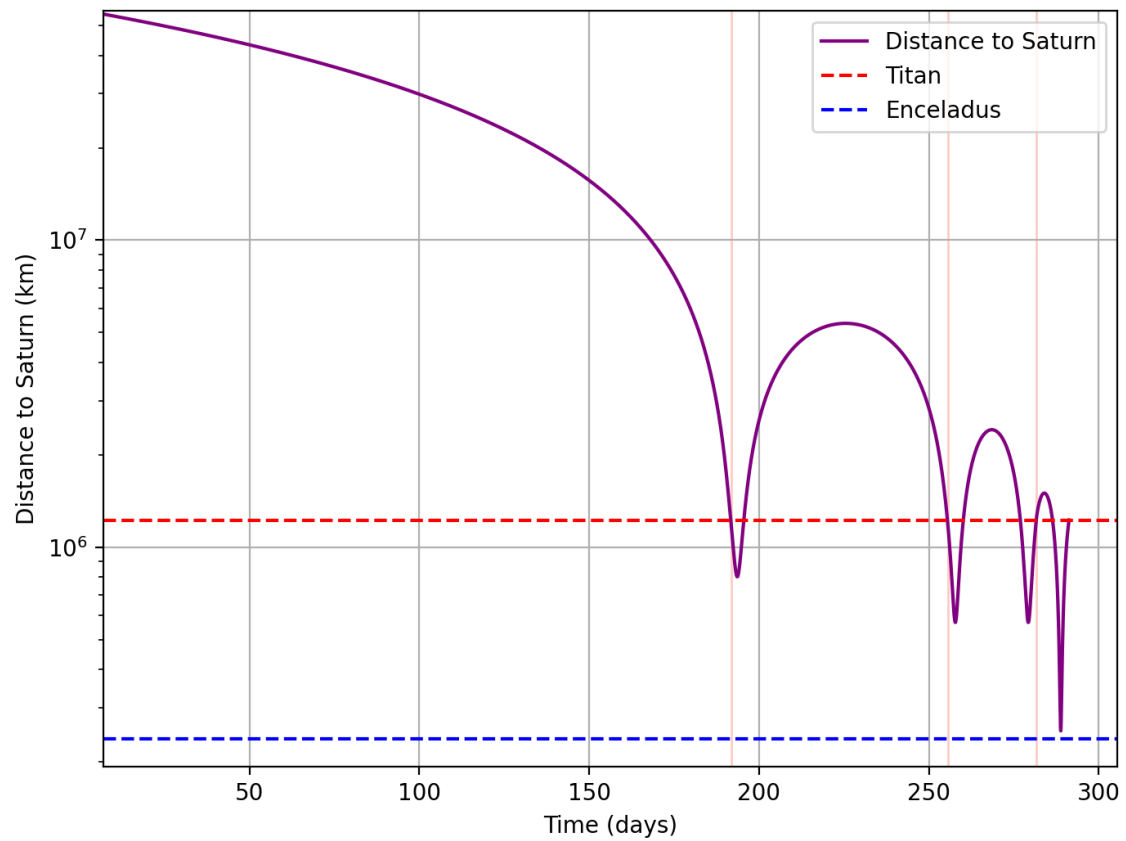


Figure 3.3: Data over time example (own work)

The Aerobrake Heights Plot, shown in Figure 3.4, illustrates the distance to Titan at the first orbit crossing that leads to subsequent rendezvous on the following orbits for different arrival velocities. The data points, marked as circles, are sorted by the first orbit's apoapsis, and a polynomial is fitted through these points to hint at possible solutions between the simulated configurations.

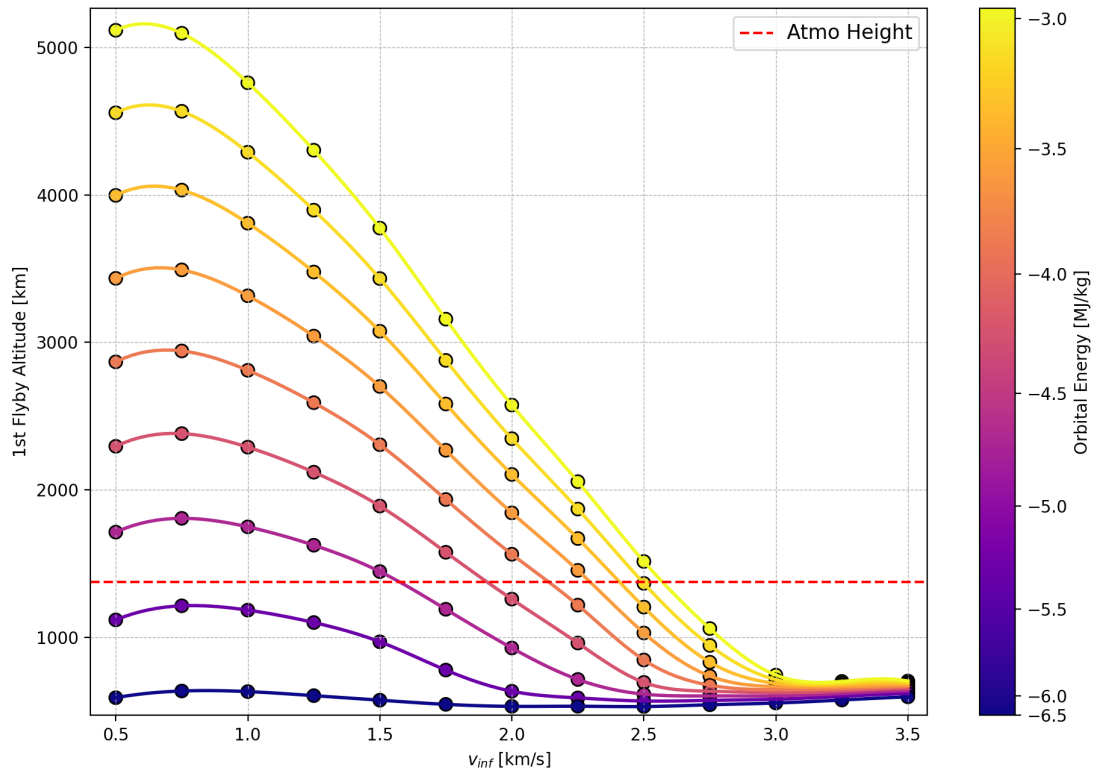


Figure 3.4: Possible altitudes for orbit capture (own work)

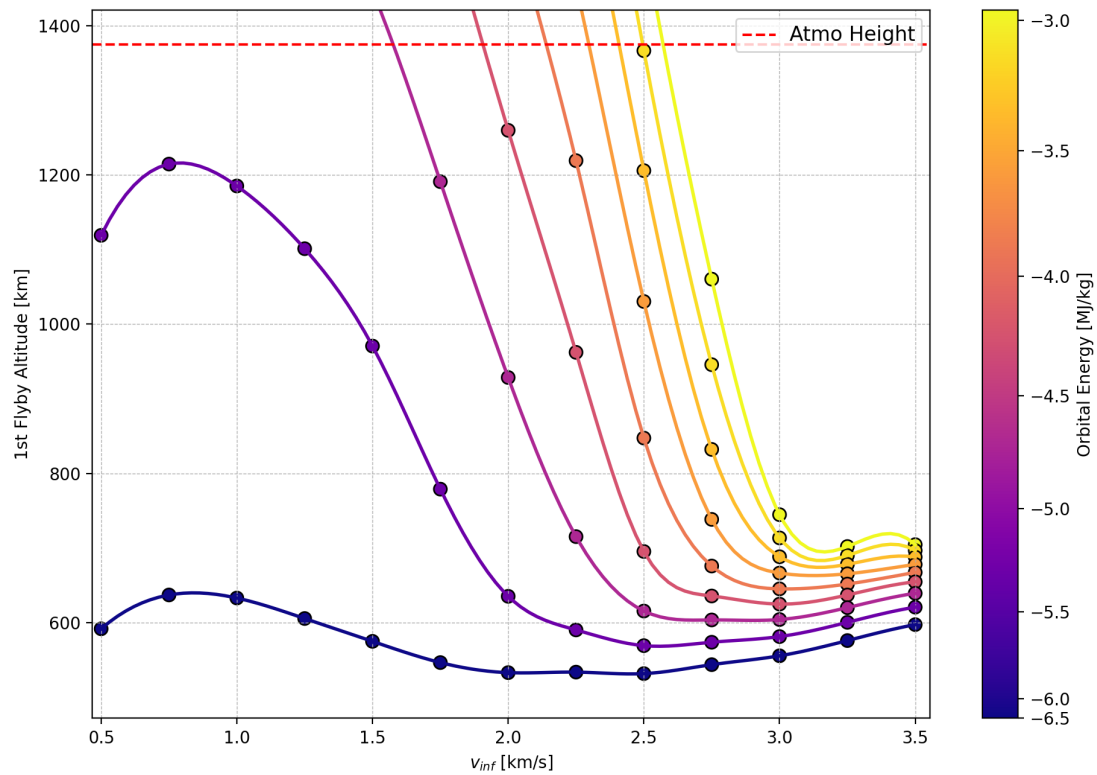


Figure 3.5: Possible aerobraking altitude for orbit capture (own work)

The Heating Rate Plot displays the heat loads generated by atmospheric drag for each configuration, as shown in Figure 3.6. The number next to each data point indicates the count of the atmospheric crossing for that entry.

NEUE HEATLOADS FIGURE

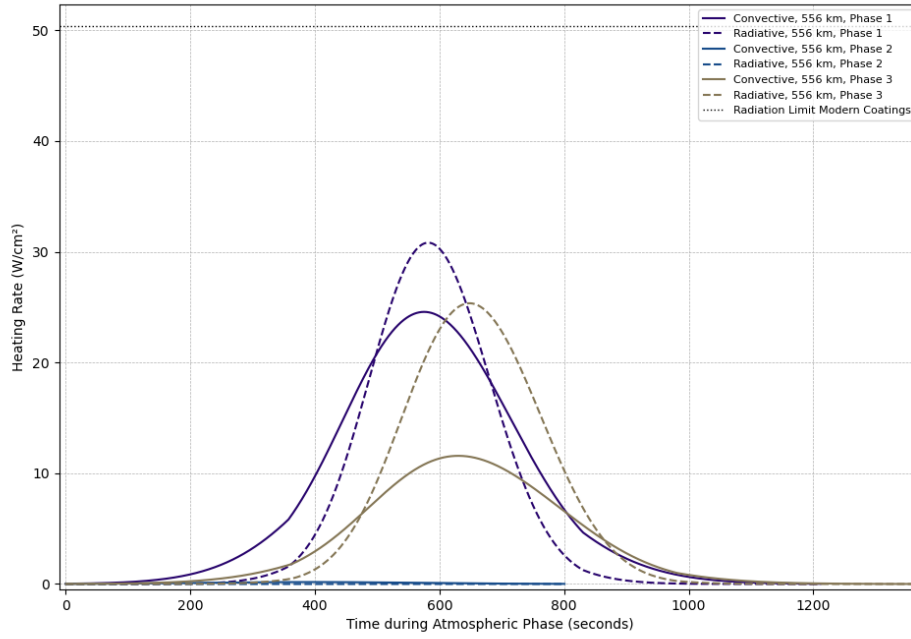


Figure 3.6: Heat loads during aerobraking (own work)

The formula for the heating rate:

$$\dot{Q} = c_q \sqrt{\rho} v^3. \quad (3.1)$$

Two types of data points are plotted:

1. Average heating rate: The total heat load for one flight through the atmosphere divided by the time the spacecraft spent in Titan's atmosphere for that crossing (marked with a circle).

$$\dot{Q}_{\text{avg}} = \frac{1}{T_{\text{atmosphere}}} \int \dot{Q}(t) dt, \quad (3.2)$$

2. Peak heating rate: The period during which the heating rate exceeds 90% of its maximum value for that crossing, ensuring that the peak heating rates remain

within the limits of the spacecraft's thermal protection system (marked with an x).

$$\dot{Q}_{peak} = \frac{1}{T_{atmosphere,j}} \int \dot{Q}_j(t) dt, \quad \text{for } j \text{ such that } Q_j \geq 0.9 Q_{max}. \quad (3.3)$$

3.2 Preparing the Program

This section of the code consists of the general planetary and orbital constants for the simulated Saturn-centered system with Titan and Enceladus found in Table 3.3. The radius, gravitational parameter, eccentricity, and semi-major axis are taken from the Horizons System by the Jet Propulsion Laboratory (Jet Propulsion Laboratory, 2022). The sphere of influence value is calculated using this data and the formula:

$$r_{SOI} = a \left(\frac{\mu_{body}}{\mu_{Saturn}} \right)^{\frac{2}{5}} \quad (3.4)$$

Furthermore, there are accuracy parameters for the simulation, unit conversion, and variables to track the spacecraft's state during the simulation being initialized.

Constant	Codename	Value	Unit
Gravitational Acceleration	g0_earth	9.80665	m/s ²
Gravitational Parameter of Saturn	mu_saturn	37931206.234	km ³ /s ²
Radius of Saturn	R_saturn	60268.0	km
Saturn's Sphere of Influence Radius	R_SOI_Saturn	54807601.125	km
Gravitational Parameter of Titan	mu_titan	8978.14	km ³ /s ²
Radius of Titan	R_titan	2575.5	km
Titan's Sphere of Influence Radius	R_SOI_titan	43321.241	km
Titan's semi-major axis	a_titan	1221870	km
Eccentricity of Titan's Orbit	e_titan	0.0288	—
Orbital Period of Titan	T_titan	15.945421	d
Gravitational Parameter of Enceladus	mu_enceladus	7.210367	km ³ /s ²
Radius of Enceladus	R_enceladus	252.3	km
Enceladus' semi-major axis	a_enceladus	238040	km
Eccentricity of Enceladus' Orbit	e_enceladus	0.0047	—
Orbital Period of Enceladus	T_enceladus	1.370218	d

Table 3.3: Planetary and orbital constants (own work)

3.3 Atmospheric Model

The atmospheric model is a standard exponential model and extends up to 1375 km altitude. The base value at the surface level is $5.38 \times 10^9 \text{ kg/km}^3$. At its farthest point, value decreases to 0.01 kg/km^3 . These values result in the following approximation:

$$\rho(h) = 5.38 \times 10^9 \text{ kg/km}^3 e^{-\frac{h}{50.904953 \text{ km}}} \quad (3.5)$$

Figure 3.7 shows the atmospheric density over altitude. The atmospheric model is based

on the data from the Huygens Atmospheric Structure Instrument. A more in-depth comparison is found in Subsection 4.2.1 Validation of the Atmospheric Model.

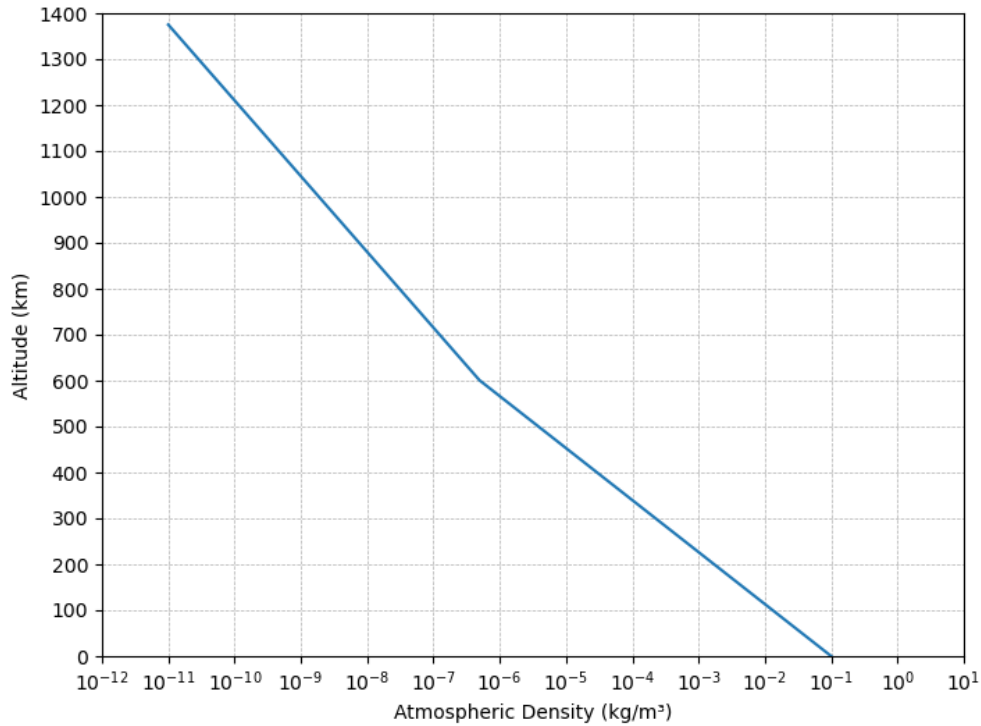


Figure 3.7: Atmospheric density model (own work)

3.4 Equations of Motion

The equations of motion section of the code defines four key functions. Three of them are helper functions:

1. `get_titan_position`: Given an initial point on Titan's orbit, this function determines Titan's position relative to Saturn at every simulation time step by solving Kepler's equation for the eccentric anomaly, then calculating the true anomaly.
2. `get_enceladus_position`: Similar to the Titan function, this calculates Enceladus'

position relative to Saturn at each simulation time step, again by solving Kepler's equation for the eccentric anomaly and then determining the true anomaly.

3. `get_initial_spacecraft_state`: This function provides the spacecraft's initial state for a given arrival velocity in both Cartesian and polar coordinates. Moreover, it also passes along the change of the radius and angle relative to Saturn at the starting point.

The fourth function is the main function, which calculates the spacecraft parameters listed in Table 3.4 at each step in the simulation. The throttle parameter describes the thrust behavior in a bang-bang control manner. The thrust system of the spacecraft in this simulation only consists of full thrust or no thrust.

Parameter	Codename	Unit
Velocity	v	km/s
Flight path angle	α	rad
Thrust angle	β	rad
Specific orbital energy	E	MJ/kg
Specific angular momentum	h	km ² /s
Semi-major axis	a	km
Eccentricity	e	—
Apoapsis	apoapsis	km
Periapsis	periapsis	km
Descending	is_descending	True/False
Throttle	throttle	0/1
Atmospheric density	atmo_density	kg/km ³
Airdrag	D	kg km/s ²

Table 3.4: Spacecraft status parameters (own work)

The main function uses the parameters of Table 3.5 to solve the following five differential equations and continuously update the state of the spacecraft:

$$\frac{dr}{dt} = \rho_r \quad (3.6)$$

$$\frac{d\varphi}{dt} = \rho_\varphi \quad (3.7)$$

$$\frac{d\rho_r}{dt} = r \rho_\varphi^2 - \frac{\mu}{r^2} - \frac{D}{m} \cos \alpha + \frac{\eta F_{Thrust}}{m} \cos \beta \quad (3.8)$$

$$\frac{d\rho_\varphi}{dt} = -2 \rho_r \rho_\varphi - \frac{D}{m} \sin \alpha + \frac{\eta F_{Thrust}}{m} \sin \beta \quad (3.9)$$

$$\frac{dm}{dt} = -\frac{\eta F_{Thrust}}{c_{eff}} \quad (3.10)$$

with the Airdrag D being calculated as:

$$D = \frac{1}{2} c_d \rho A v^2 \quad (3.11)$$

Parameter	Codename	Formula symbol	Unit
Spacecrafts Radial Position	r	r	km
Spacecrafts Angular Position	phi	φ	rad
Change in Spacecrafts Radial Position	rhorr	ρ_r	km/s
Change in Spacecrafts Angular Position	rhophi	ρ_φ	rad/s
Gravitational Parameter	mu	μ	km ³ /s ²
Mass	m	m	kg
Flight path angle	alpha	α	rad
Thrust direction angle	beta	β	rad
Throttle	throttle	η	0/1
Thrust	thrust	F_{Thrust}	kN
Exhaust velocity	ceff	c_{eff}	km/s
Airdrag	D	D	kg km/s ²
Drag Coefficient	c_d	c_d	—
Atmospheric density	atmo_density	ρ	kg/km ³
Cross-Sectional Area	A	A	km ²
Velocity	v	v	km/s

Table 3.5: Equations of motion formula symbols (own work)

There are different approaches to simulating the gravitational influences of multiple bodies on a spacecraft. A commonly used method is the patched conic approximation, which applies solely the force of the dominant body to the spacecraft. This requires clearly separated regions and mission phases. In this thesis, this means that for most of the mission, only Saturn's gravity is applied to the spacecraft. When the spacecraft enters Titan's sphere of influence, Saturn's gravity will be neglected and Titan's gravity will be applied. While this approach simplifies calculations, since only one gravitational force is considered at a time, it can lead to unstable results near the boundaries of Titan's sphere of influence. This instability is especially problematic when Titan and the spacecraft

move in similar directions, causing the spacecraft to leave and re-enter Titan's sphere of influence multiple times, because the direction and magnitude of the force applied to the spacecraft changes rapidly. For the precise maneuver of this mission, constantly applying both gravity forces matches real-world dynamics more closely. Therefore, the gravitational effects of both Titan and Saturn are applied simultaneously to the spacecraft.

3.5 Launch Delay Calculation

The Launch Delay Calculation aims to find a launch time that results in the spacecraft passing in front of Titan at a specific altitude out of a database. Further explanation on this first crossing distance and how it is generated is found in Section 3.6 Adjust Aerobraking Altitude. If the configuration of arrival velocity and first orbit crossing altitude already has resulted in valid results, the function will use the launch time from the database. If the database has no entry for the configuration, the launch delay calculation function starts an iteration process that follows a distinct logic. The function first simulates the trajectory of the spacecraft until it crosses Titan's orbit the first time. It then checks for the angular difference between the spacecraft and Titan at that time. A positive angular difference means that the spacecraft is ahead of Titan along its orbit. Hence, the launch delay needs to be increased so that Titan has more time to travel toward the crossing point. A negative angular difference thus results in a decrease in launch delay. The adjustment in launch time is calculated based on the angular difference $\Delta\varphi$ divided by the average angular velocity of Titan along its orbit Ω_{avg} . Since Titan's orbit has a low eccentricity of 0.0288, its angular velocity is relatively consistent, resulting in a fast angular adjustment until a threshold of up to 2° is reached. Within that, the function switches to an adjustment based on the distance in altitude to the first crossing target in km. The adjustment in launch delay now uses a coefficient of around 0.187 and multiplies it with the distance to target altitude. The coefficient varies between 0.185 and 0.2 depending on the arrival velocity. However, the general logic of increasing the delay when the spacecraft's current altitude is higher than the target altitude and vice versa stays the same. The entire iteration process works with error-based adjustments, meaning that larger errors lead to larger adjustments and therefore result in fast convergence. The full logic of the iteration process is illustrated in a flow diagram in Figure 3.8.

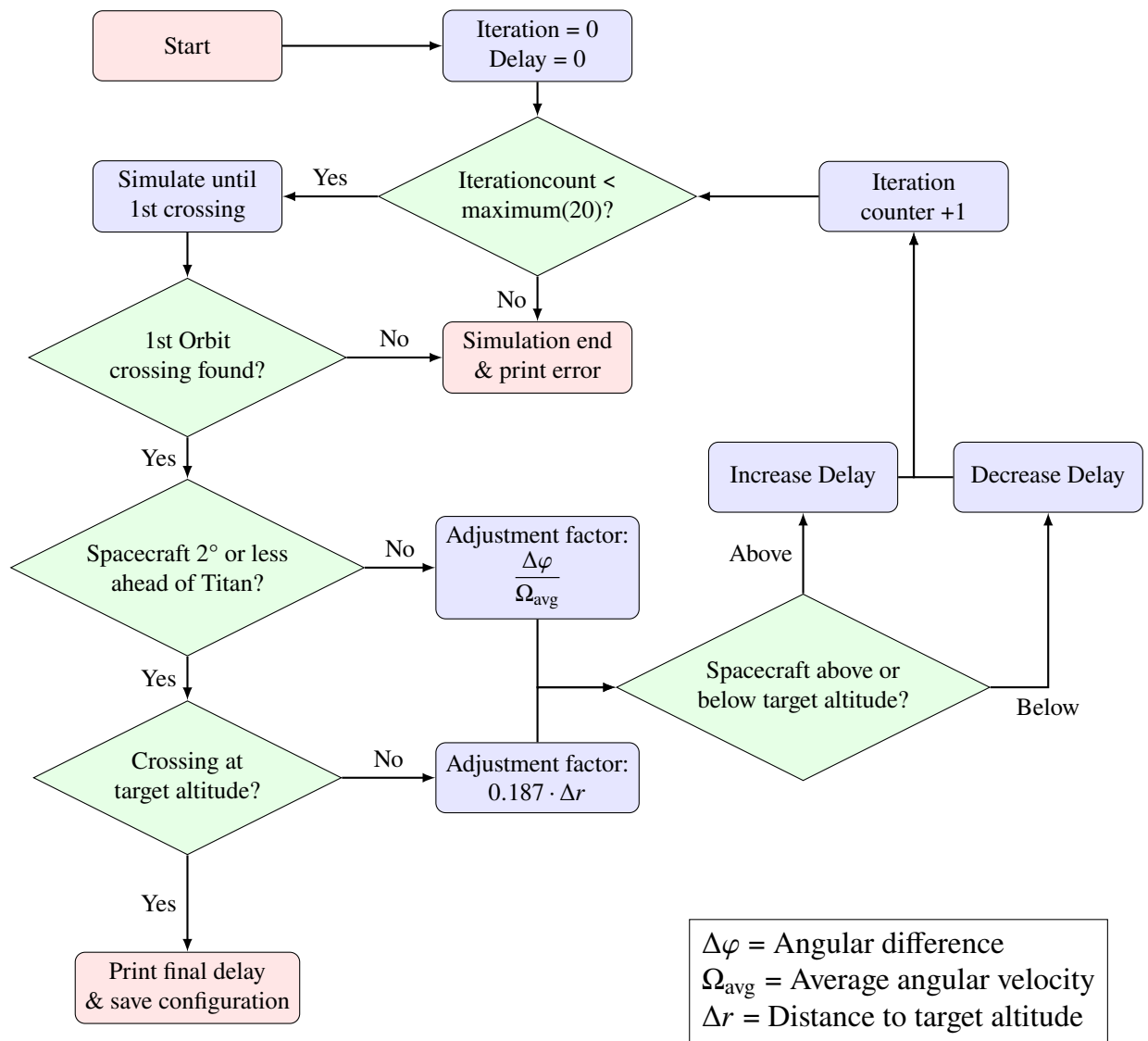


Figure 3.8: Logic of the launch delay iteration (own work)

An example of how the iteration process looks for values that are not yet listed in the database can be found in Figure 3.9.

```
[0.50 km/s] Delay-Iteration 1 for r_p = 3740.911812 km: Spacecraft is 20.85 degrees away from Titan
Adjusting delay by 20.58 hours to 0.8575286511 days.
[0.50 km/s] Delay-Iteration 2 for r_p = 3740.91 km: Spacecraft is 1912.1872 km too far ahead of Titan.
Increasing delay by 5.53 minutes to 0.86136712 days.
[0.50 km/s] Delay-Iteration 3 for r_p = 3740.91 km: Spacecraft is 0.6636 km too far ahead of Titan.
Increasing delay by 0.12 seconds to 0.86136847 days.
ITERATION DONE FOR [0.50 km/s] | C1: 3740.91 km | C2: 1411898.83 km | C3: 2261.84 km | DELAY: 0.86136847 days
```

Figure 3.9: Delay iteration example (own work)

Once a suitable delay time is found for the chosen configuration, the program moves to Section 3.6 Adjust Aerobraking Altitude

3.6 Adjust Aerobraking Altitude

This function utilizes the launch delay found in Section 3.5 Launch Delay Calculation to simulate all configurations with their respective launch delays and first crossing distances over an extended duration. The exact simulation duration is set by the user in Section 3.1 User Input and is limited either by the maximum number of Titan orbit crossings or the maximum integration time. During the simulation, the function tracks the spacecraft's relative position to Titan for every crossing of Titan's orbit. Its main goal is to adjust the first crossing distance to a value that leads to subsequent rendezvous of the spacecraft and Titan. More specifically, the database contains crossing values for as many rendezvous as the user requires. In practice, four Titan flybys have proven effective for lowering the spacecraft's orbit closely to Enceladus. Table 3.6 illustrates the structure of a typical database entry and Figure 3.10 is the matching trajectory for this configuration.

Arrival Velocity	1st Flyby	2nd Flyby	3rd Flyby	Launch Delay
3.0 km/s	555.56 km	1015.88 km	584.52 km	7.12066 days

Table 3.6: Database entry (own work)

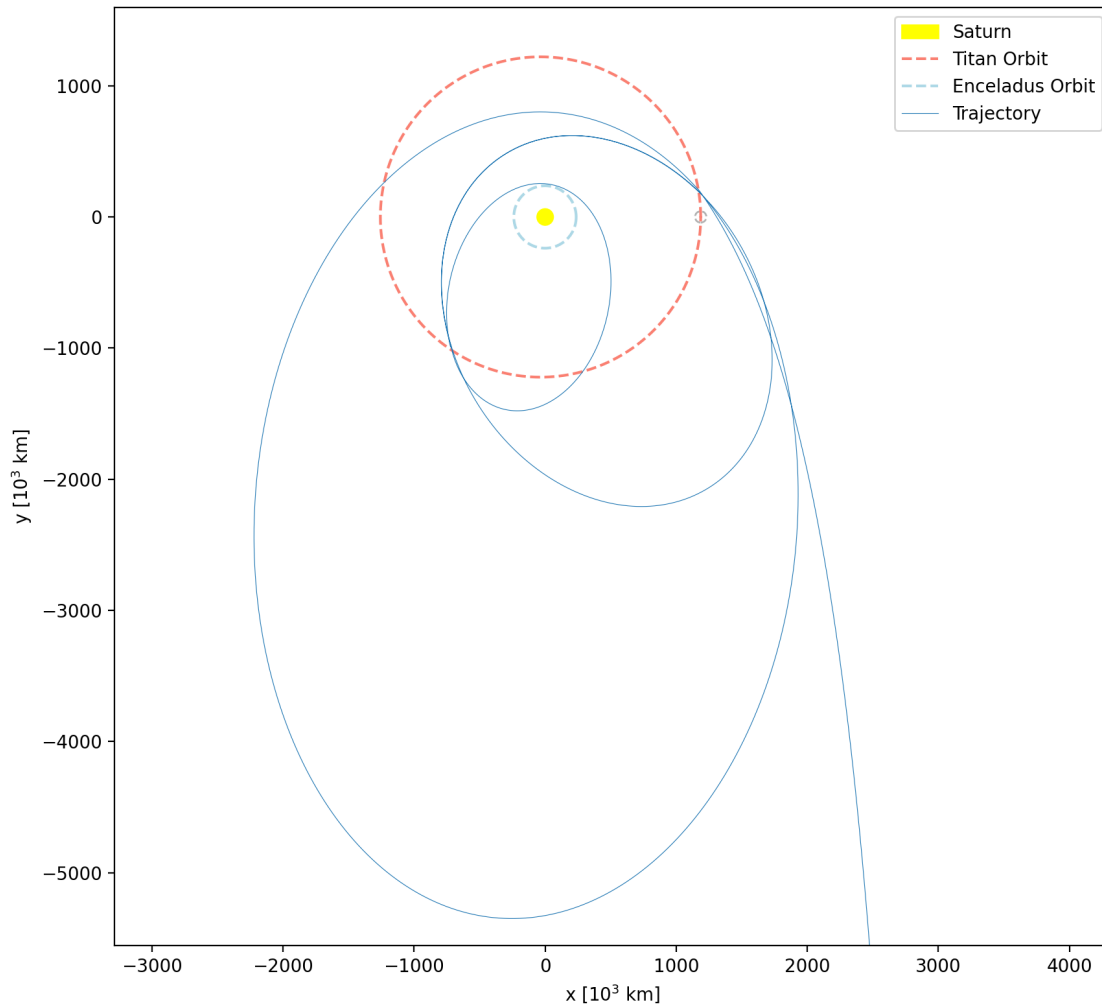


Figure 3.10: Database entry trajectory (own work)

When creating a new entry in the database, the function first calls the Launch Delay function to get a valid launch time for the initial flyby distance. Since the launch delay calculation ensures the first crossing distance is at the desired altitude, the function skips that check and examines the spacecraft's position relative to Titan at the second flyby.

The adjust aerobrake height function then, similarly to the adjust launch delay function, first checks for the angular difference between the spacecraft and Titan at the second crossing. There are two differences to the launch time function:

1. The adjustment factor is significantly smaller. A change of 0.1 km in the first crossing altitude results in a difference of 900 km at the second crossing and 57°

at the 3rd crossing. The adjustment coefficient that gets multiplied with the error decreases by a factor of approximately 1500–3000 for every consecutive crossing. So the coefficient of 0.187 from Section 3.5 Launch Delay Calculation becomes 0.0007 for the second crossing and 0.000000045 for the third crossing.

2. For even crossings the adjustment is inverted. The apoapsis of the following orbit changes significantly with the altitude of every flyby. Hence, lowering the first orbit crossing distance will result in a bigger energy loss due to Titan’s gravity and atmospheric drag. This leads to a smaller orbit and a shorter orbital period for the spacecraft. Thus, the spacecraft will be at the rendezvous point faster than before, resulting in an increase in distance to Titan at the second orbital crossing.

After each adjustment to the first crossing distance, the Launch Delay function is invoked again to ensure the desired changed altitude is met. This iterative process continues until the second flyby distance aligns with the user-specified value and the given tolerance of 1 km. The same procedure is then applied to the third flyby distance, continually readjusting the first crossing distance and launch delay until the third flyby also meets the required distance with a given tolerance of 100 km, as well as for the fourth flyby with a tolerance of 500 km. The tolerances are adjustable to the needs of the user. It is recommended to increase the tolerances with every consecutive crossing, as the required accuracy to meet the tolerances increases with every consecutive crossing, resulting in large computation times when looking for small tolerances at the fourth crossing or beyond, due to very small simulation step sizes and error margins. Upon completion, the program prints a summary, as shown in Figure 3.11. This print statement contains the arrival velocity, all flyby distances, the final periapsis distance to Enceladus, and the launch delay used.

```
ITERATION DONE FOR [3.00 km/s] | C1: 555.5586265156 km | C3: 1015.8802680677 km | C6: 583.5230166957 km  
LAST PERIAPSIS DISTANCE TO ENCELADUS: 13760 km | DELAY: 7.12066324864928 days
```

Figure 3.11: Final print statement (own work)

Since the function continuously changes the first crossing distance and no thrust maneuvers are taking place during the mission, the user-given values for the crossings before the last crossing will get changed. For example, once the second crossing distance is within the tolerance and the function starts adjusting for the third crossing, the second crossing altitude will change, as it is dependent solely on the first crossing altitude, which

is being adjusted. However, the change in orbital energy by enabling a third crossing significantly outweighs any potential losses due to an increase in the second crossing altitude. Taking Figure 3.12 shows that if the user specified the second flyby C3 to be at 413 km originally, the program will search for a first crossing altitude C1 that results in C3 being at 413 km. To achieve a third flyby at C6 = 583 km, however, C1 will need to be adjusted again and, as a result, C3 increases to 583 km. The last periapsis distance to Enceladus, however, is almost 150,000 km lower by enabling the C6 crossing to take place than by having C3 set to 413 km as shown in Figure 3.11.

```
ITERATION DONE FOR [3.00 km/s] | C1: 555.5609947448 km | C3: 413.2046571134 km  
LAST PERIAPSIS DISTANCE TO ENCELADUS: 150830 km | DELAY: 7.12066324637848 days
```

Figure 3.12: Two vs. three flyby comparison (own work)

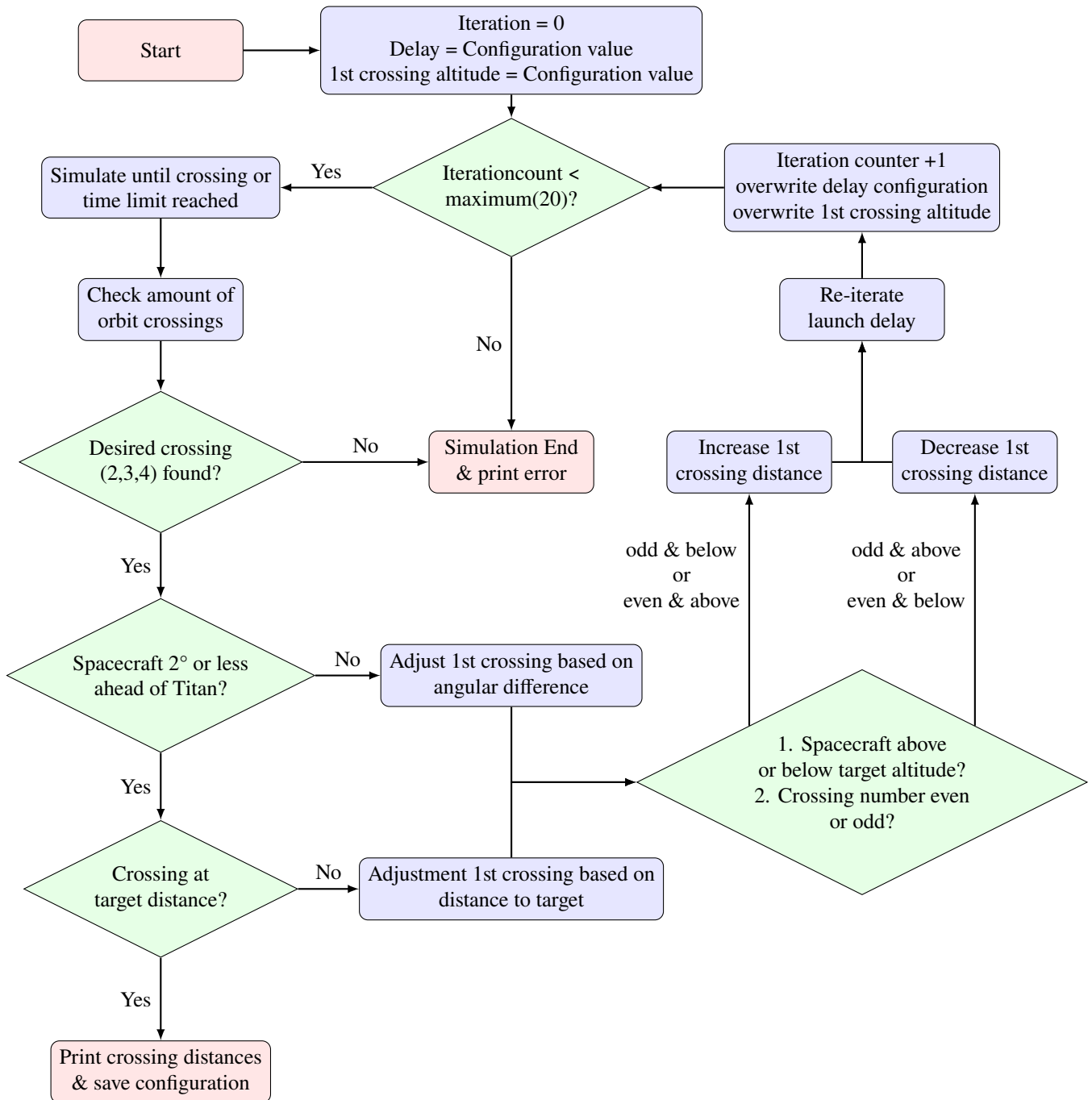


Figure 3.13: Logic of the aerobrake height iteration (own work)

4 Results

4.1 Simulation Results

Analyzing possible flyby altitudes to stay within the Saturnian system results in finding multiple possible flyby altitudes for every hyperbolic excess velocity up to 3.5 km/s. The reduction in orbital energy for the first flyby at Titan needs to be strong enough to ensure that the spacecraft does not leave Saturn's sphere of influence. The apoapsis r_a has to be smaller than the radius of the sphere of influence from Saturn r_{SOI_Saturn} . A high first flyby altitude results in a lower gravitational pull of Titan and, if it's within the atmosphere, a lower atmospheric density and therefore a lower experienced air drag. As shown in Subsection 2.1.2 Gravity Assist and Aerobraking the gravitational pull decreases with the square of the distance between them and the force of air drag is directly proportional to the density of the atmosphere. According to the atmosphere model shown in Figure 3.7 the density decreases exponentially with altitude. Hence, the experienced force slowing down the spacecraft is heavily dependent on the altitude to Titan. It is therefore to be expected that the higher the first flyby altitude, the larger is the first apoapsis and with that the orbital period of the first orbit. This assumption is proven to be correct by Figure 4.1

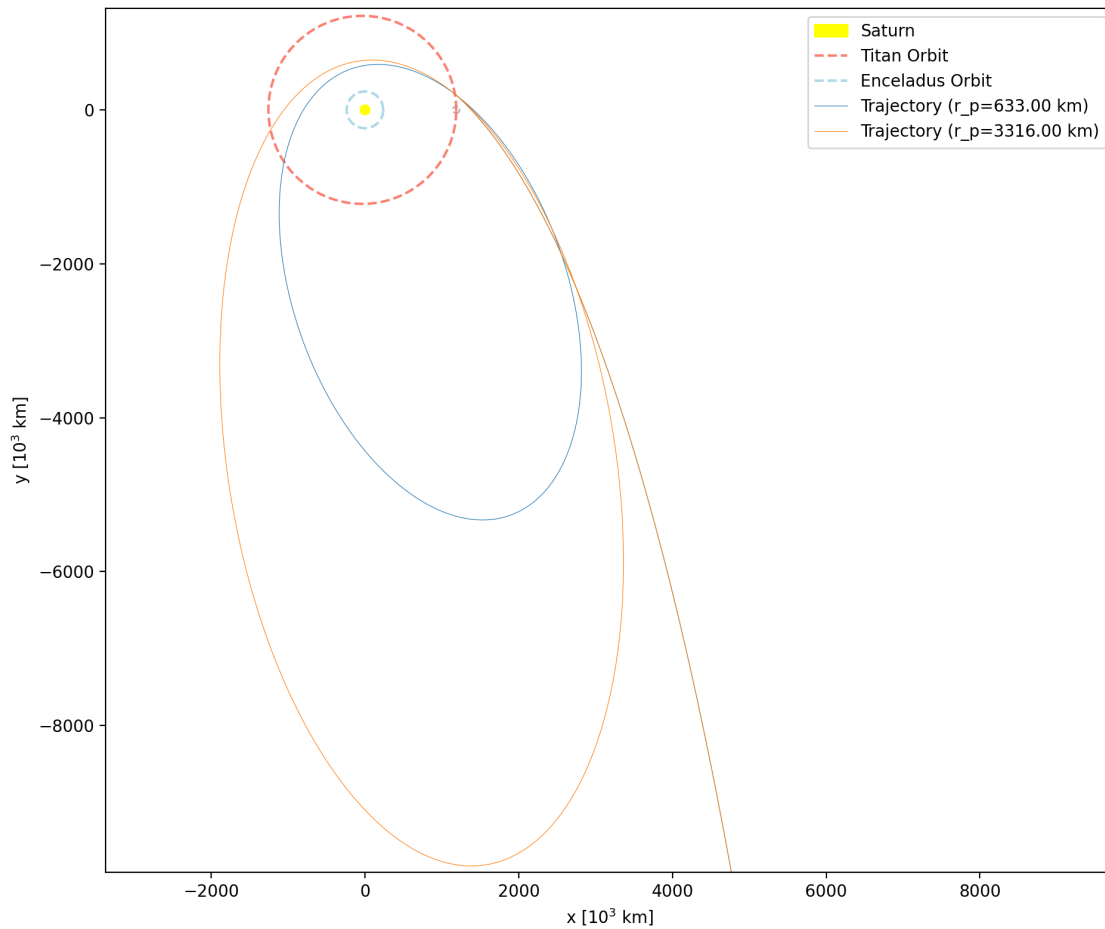


Figure 4.1: Different flyby altitude trajectory comparison (own work)

Therefore, when optimizing for time of flight the mission analysis goal is to use the lowest flyby altitude possible.

There is a limit on how low the first aerobraking altitude can be. This limit can be decided by different factors. Firstly, it is important to consider a safety margin that allows for small deviations between the simulation and the real mission later. Checking for the lowest possible aerobraking altitude that doesn't result in a crash into Titan due to the combined forces of gravity and aerodynamic drag delivers the result from Figure 4.2

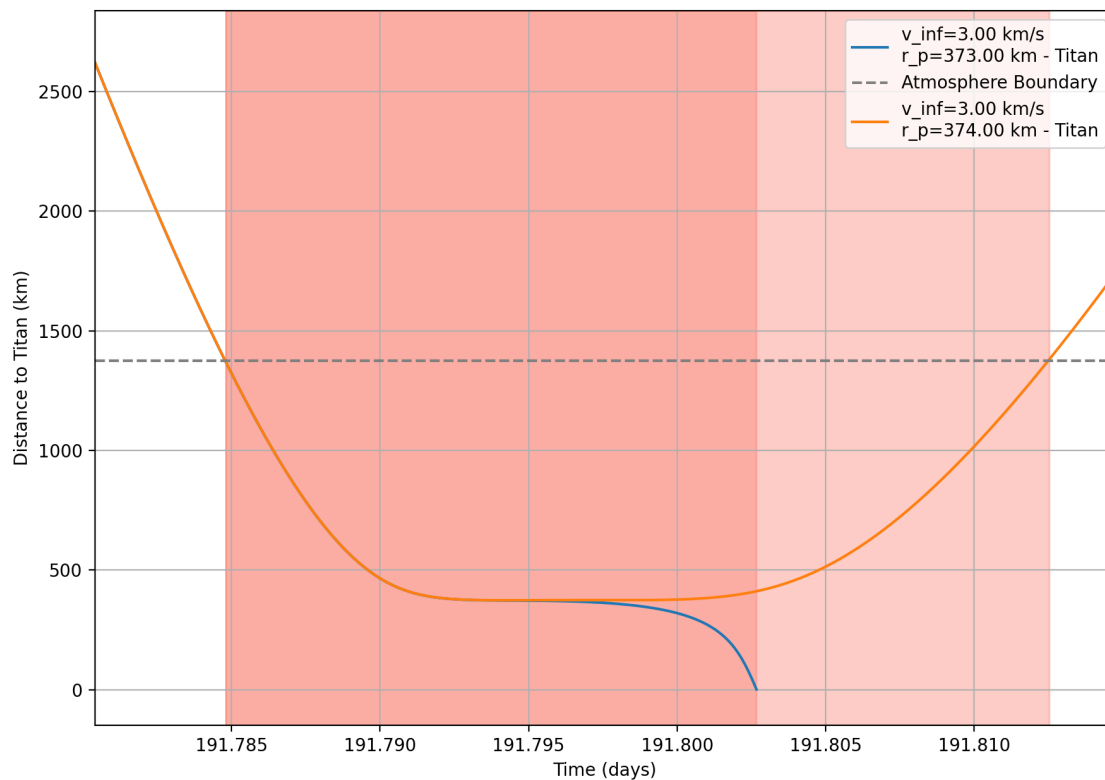


Figure 4.2: Lowest possible aerobraking altitude (own work)

Applying a safety margin of 10% to that altitude to ensure a large enough space between simulated mission failure therefore limits the aerobraking maneuver to a minimal first altitude of 411.4 km.

Another constraint on the lowest possible altitude is the heat generated by friction between the spacecraft's cross-sectional area and the atmosphere. As discussed in Section 3.1 there are three primary ways for managing this thermal load. Consequentially, depending on mission constraints in terms of available mass to dissipate heat there is another limiting factor. However, the limit of the ablative heat dissipation method essentially defines the spacecraft's overall thermal tolerance, as there is currently no superior heat shielding technology available. For the spacecraft examined in this thesis, which has an aeroshell diameter of 3.75 m, no configuration exceeds the heat loads that PICA can withstand. ?? shows the highest heat loads examined in this thesis.

INSERT HIGHEST THERMAL LOAD OF THESIS

Considering that PICA can withstand 20 times the thermal load than radiative coatings the pure heat constraint can be neglected for the purposes of this study (Laub, 2012). The only remaining concern is whether a radiative thermal protection system will be sufficient or if a heat sink or ablative approach is necessary.

4.2 Validation of Results

To ensure the reliability of the simulation results, the program developed in this thesis was validated by comparing its outputs with previously published studies and flight data in four critical areas:

1. The atmospheric model
2. Gravitational forces
3. Aerodynamic forces
4. Aerothermodynamics

Furthermore, this work builds on a previously developed martian aerobraking program, which demonstrated less than a 1% error in predicting apoapsis changes when compared with data from the ESA ExoMars mission from 2017, using equivalent initial conditions (Kalscheuer, 2025).

4.2.1 Validation of the Atmospheric Model

As described in Section 3.3 Atmospheric Model, the atmospheric model is based on data collected by the Huygens Atmospheric Structure Instrument (HASI) during the Cassini-Huygens mission from 1997 until 2017. HASI recorded Titan's atmospheric properties, such as altitude, temperature, pressure, and density, during its descent on January 14, 2005. Figure 4.3 illustrates how HASI continuously tracked these parameters during the descent.

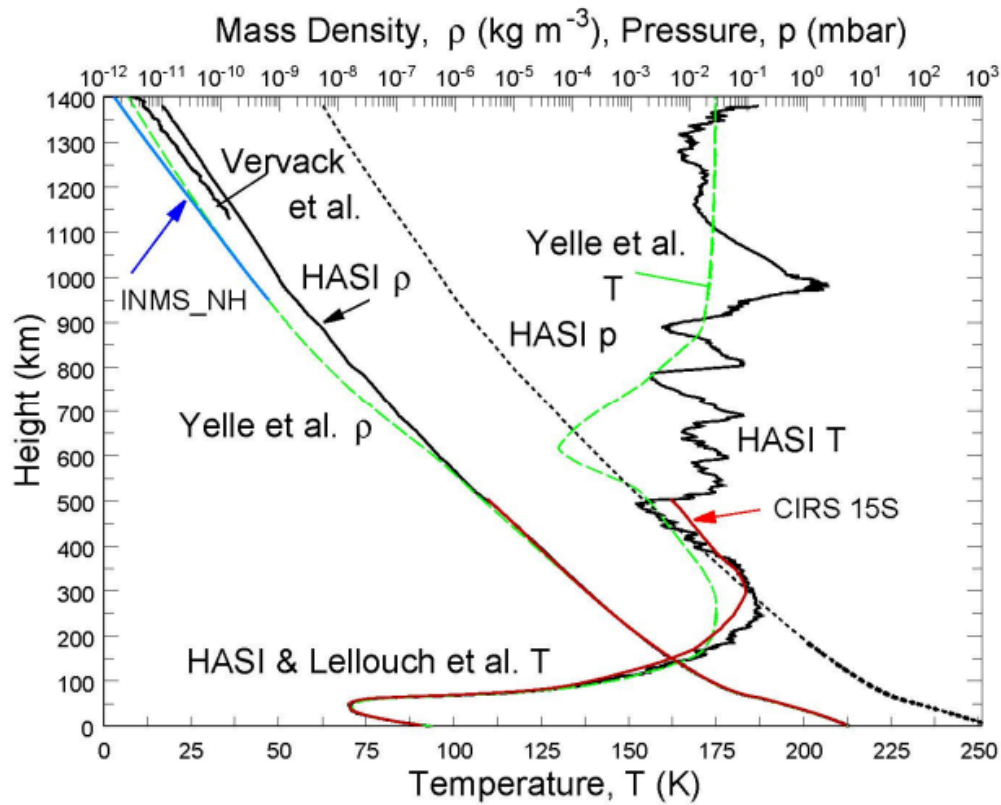


Figure 4.3: Atmospheric data from HASI (Jet Propulsion Laboratory, 2019)

Figure 4.4 compares the model (blue line) with the HASI data (red dots) in a single plot.

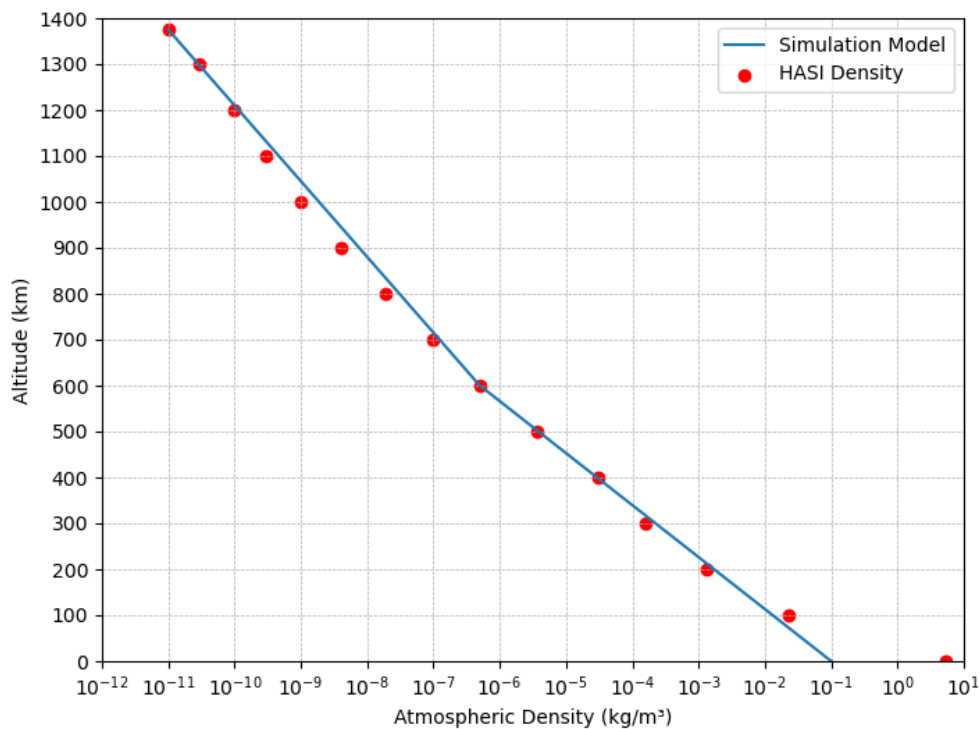


Figure 4.4: Comparison of the atmospheric model and HASI data (own work)

Notably, the model aligns precisely with the data at two key altitudes: at the outer boundary of the atmosphere (1375 km) and at 600 km. Although the model does not perfectly reproduce all measured values, it is slightly denser than the observations. This conservative bias ensures that the simulation does not underestimate atmospheric density, thereby reducing the risk of mission failure.

The model is structured in two sections to account for different density gradients in the atmosphere. Below 600 km, the increase in density is steeper than above this altitude. It should also be noted that below 200 km the modeled density begins to fall below the observed values. However, any aerobraking maneuver below 200 km is impractical due to excessive heat loads at that altitude and the need for a safe distance to Titan's surface.

Since HASI is the only source of atmospheric data for Titan, it is unclear whether the density varies with solar activity in a manner similar to the Martian atmosphere.

Therefore, a conservative approach is advisable to ensure mission safety. Short thrust maneuvers allow necessary adjustments for any discrepancies between the spacecraft's actual trajectory and the simulation.

In summary, although the model is relatively simple, it is accurate enough to yield reliable results for mission analysis and planning, ultimately helping to ensure the safety of the mission, while lowering fuel requirements.

4.2.2 Validation of Gravitational Forces

The simulation's treatment of gravitational forces was validated by comparison with an Aerocapture Design Study for a Titan Polar Orbiter (Nixon et al., 2016). To ensure comparable results the parameters of the program are adjusted to match the ones from the study. Namely the hyperbolic arrival velocity is set to $v_{\text{inf}} = 4.34$ km/s as Figure 4.5 from the study shows. Moreover, the B-plane-offset is set that the trajectory intersects with Titans Orbit at the same point as shown in Figure 4.6 from the paper.

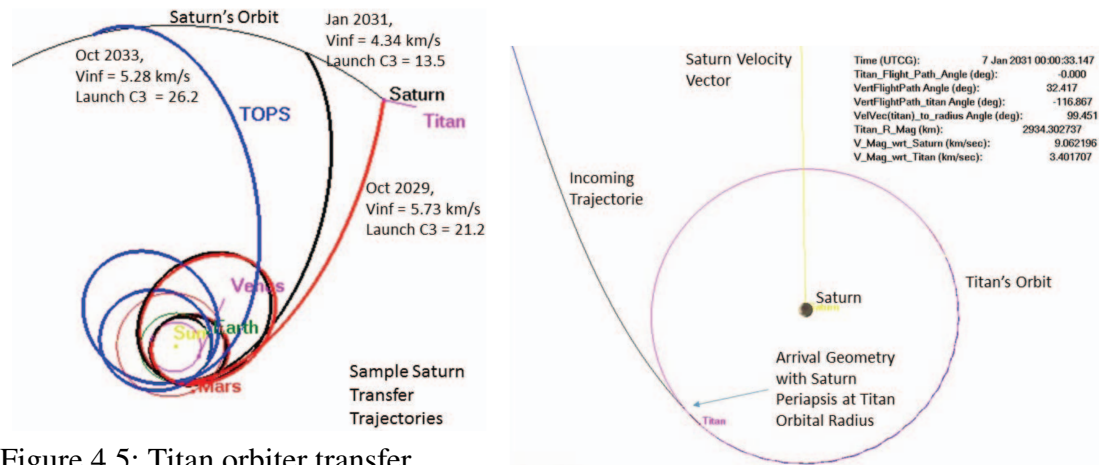


Figure 4.5: Titan orbiter transfer

trajectory (Nixon et al., 2016)

Figure 4.6: Titan orbiter arrival trajectory (Nixon et al., 2016)

At the point of intersection, the reference study reports a spacecraft velocity relative to Saturn of 9.062196 km/s, while the simulation yields 9.062 km/s as shown in Figure 4.7

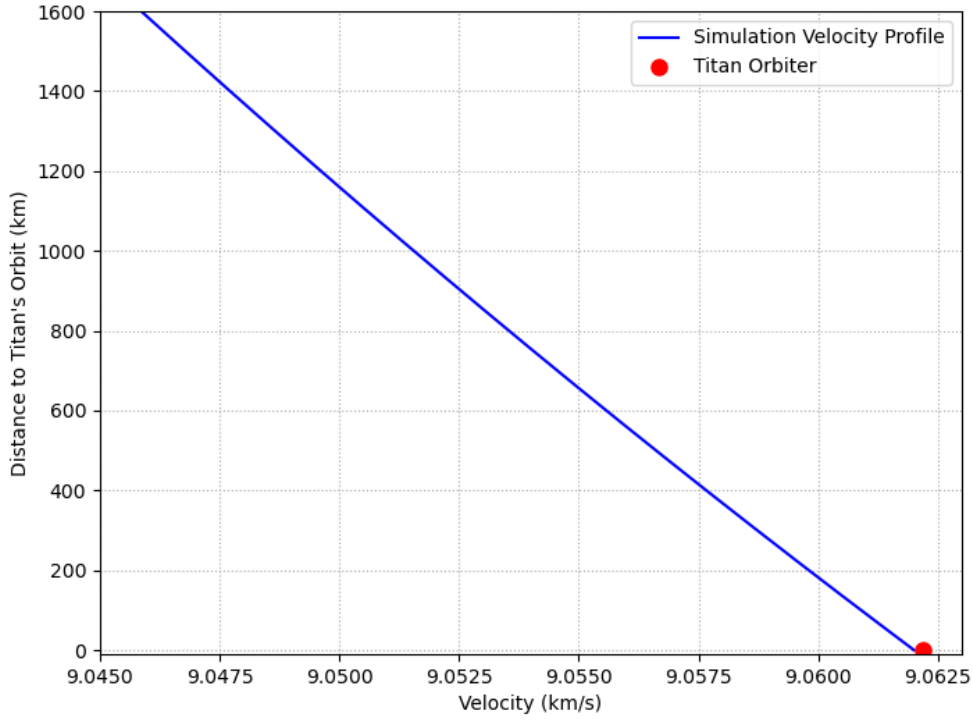


Figure 4.7: Spacecraft velocity when reaching Titan (own work)

This corresponds to an absolute error of 0.000196 km/s and a relative error of:

$$\text{Error}_{\text{rel}} = 1 - \frac{9.062 \text{ km/s}}{9.062196 \text{ km/s}} = 0.002163 \% \quad (4.1)$$

Thus, demonstrating excellent agreement between the simulation and the reference study.

4.2.3 Validation of Aerodynamic Forces

Validation of the aerodynamic forces was achieved by comparing the simulation results with those presented in NASA's study "Aerocapture System Analysis for a Titan Mission" (Langley Research Center, 2006). Although the NASA paper primarily addresses

aerocapture, its findings on atmospheric interaction across the full altitude range are directly applicable for cross-validation. The study identifies an optimal entry corridor at an angle of $36.8^\circ \pm 1.0^\circ$ and uses an aeroshell with a diameter of 3.75 m yielding a surface area of 11.04466 m^2 . The aerocapture results from NASA are based on the Hybrid Predictor-corrector Aerocapture Scheme (HYPAS) algorithm. The algorithm has previously been tested extensively in simulations for aerocapture at Earth and Mars, and this study was the first one to apply it for Titan. So the HYPAS algorithm gives reliable results that have been tested and validated by NASA before for multiple celestial bodies (Langley Research Center, 2006). Figure 4.8 shows the velocity of the spacecraft at different altitudes during the aerocapture maneuver.

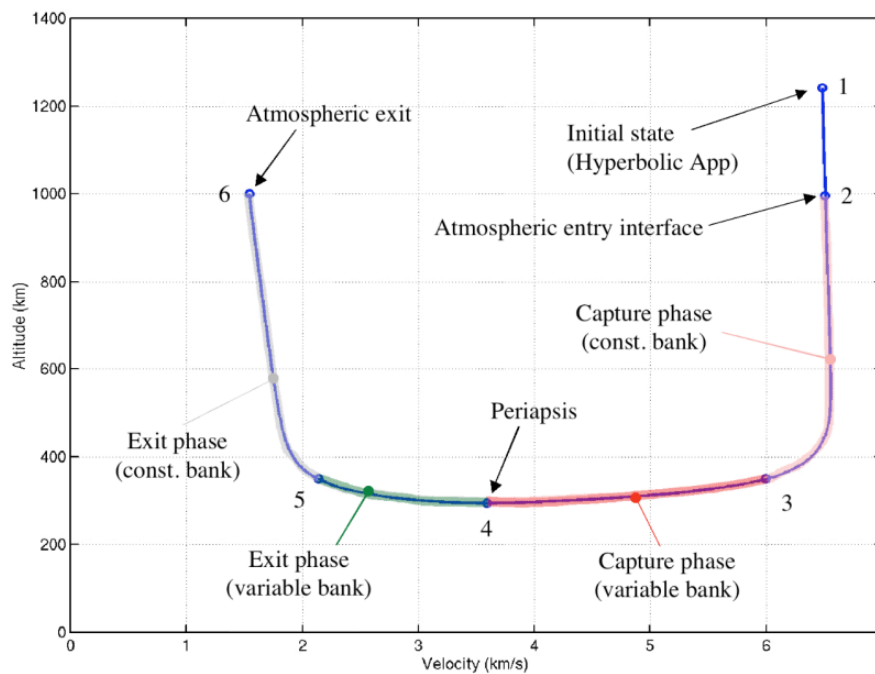


Figure 4.8: HYPAS aerocapture guide (Way et al., 2003)

In order to perform a direct comparison, the simulation was adjusted to match the initial conditions used in the NASA study. Specifically, the atmospheric height was set to 1000 km, the entry flight path angle to 36.8° , and the entry speed to 6.5 km/s relative to Titan. These values were chosen because they are the parameters used by the study to define the initial spacecraft state for the maneuver. With these parameters, the simulation produced a velocity profile during the aerocapture maneuver that can be directly compared to the

one from the study. As shown in Figure 4.9, the simulation closely matches the HYPAS results, with deviations limited to approximately 20 km periapsis altitude and 0.1 km/s velocity. These minor differences are likely due to variations in the atmospheric model used, but overall, the strong agreement confirms that the simulation accurately captures the aerodynamic forces during the maneuver.

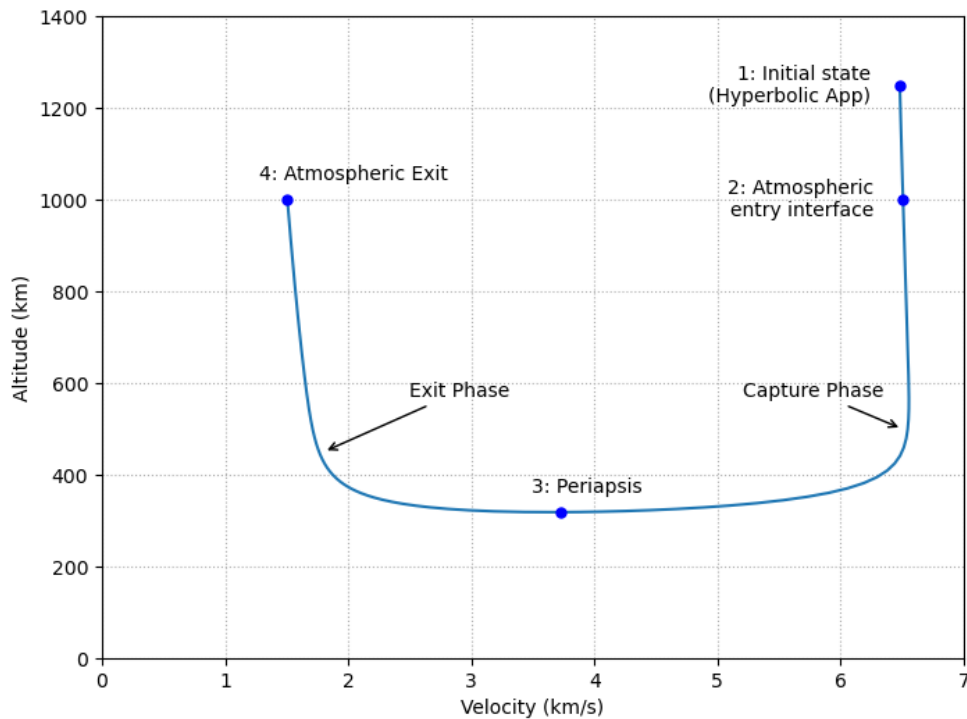


Figure 4.9: Program aerocapture guide (own work)

This validation reinforces the reliability of the model in predicting the behavior of the spacecraft under aerodynamic forces, thereby supporting its applicability for mission planning and analysis.

4.2.4 Validation of Aerothermodynamics

To ensure the heat loads given by the program model are realistic a comparison to a NASA study is done. The green graph in Figure 4.10 represents the radiative and convective heat loads of the aerocapture maneuver for the nominal atmosphere at 6.5 km/s.

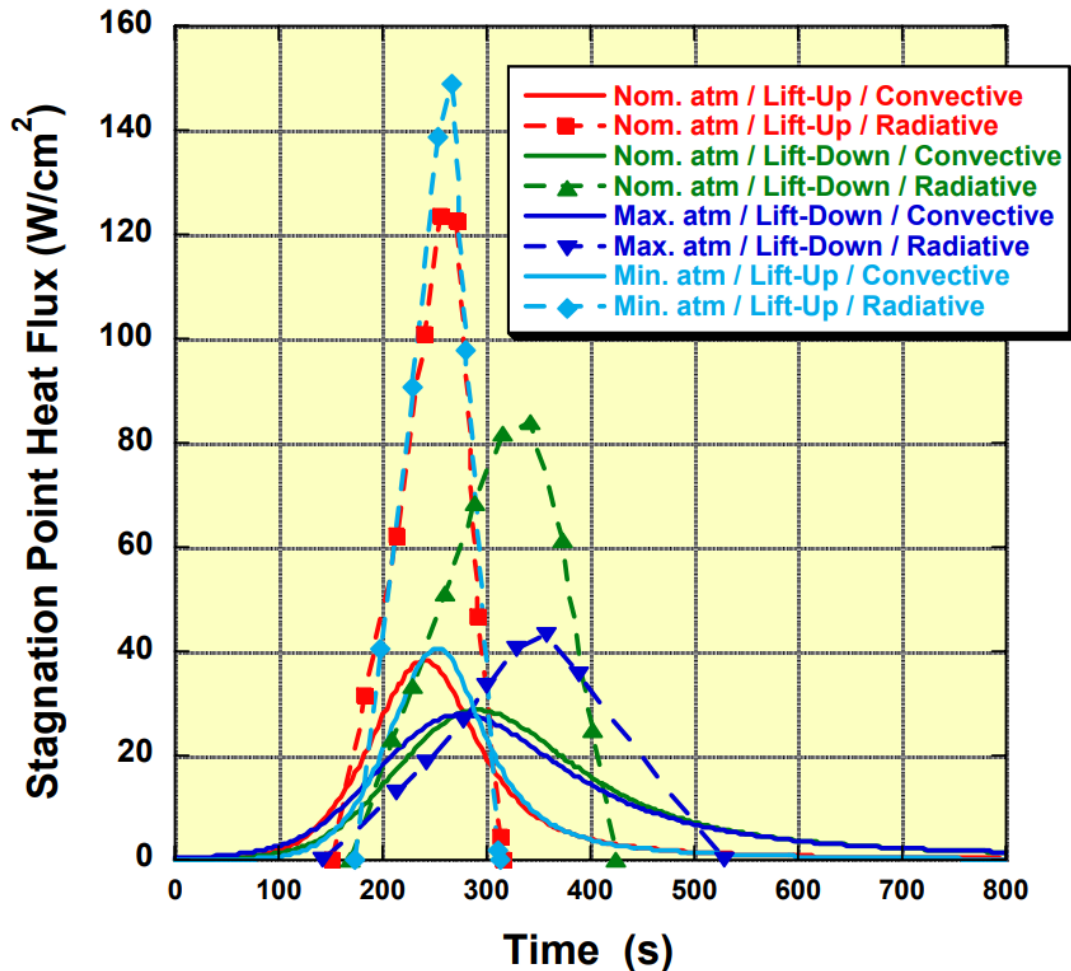


Figure 4.10: Heatflux during aerocapture NASA study (Laub, 2012)

In comparison Figure 4.11 shows the heat loads simulated by the model used in this thesis. Although there is a noticeable difference, the reason for that is most likely a difference in atmospheric model and trajectory. A visual comparison of the atmospheric

model and the altitude and velocity during the aerocapture is found in Section A.3. The simulated heat loads are within a margin of X % which allows to confidently say that the simulated heat loads for the results do match the respective trajectory and atmospheric model.

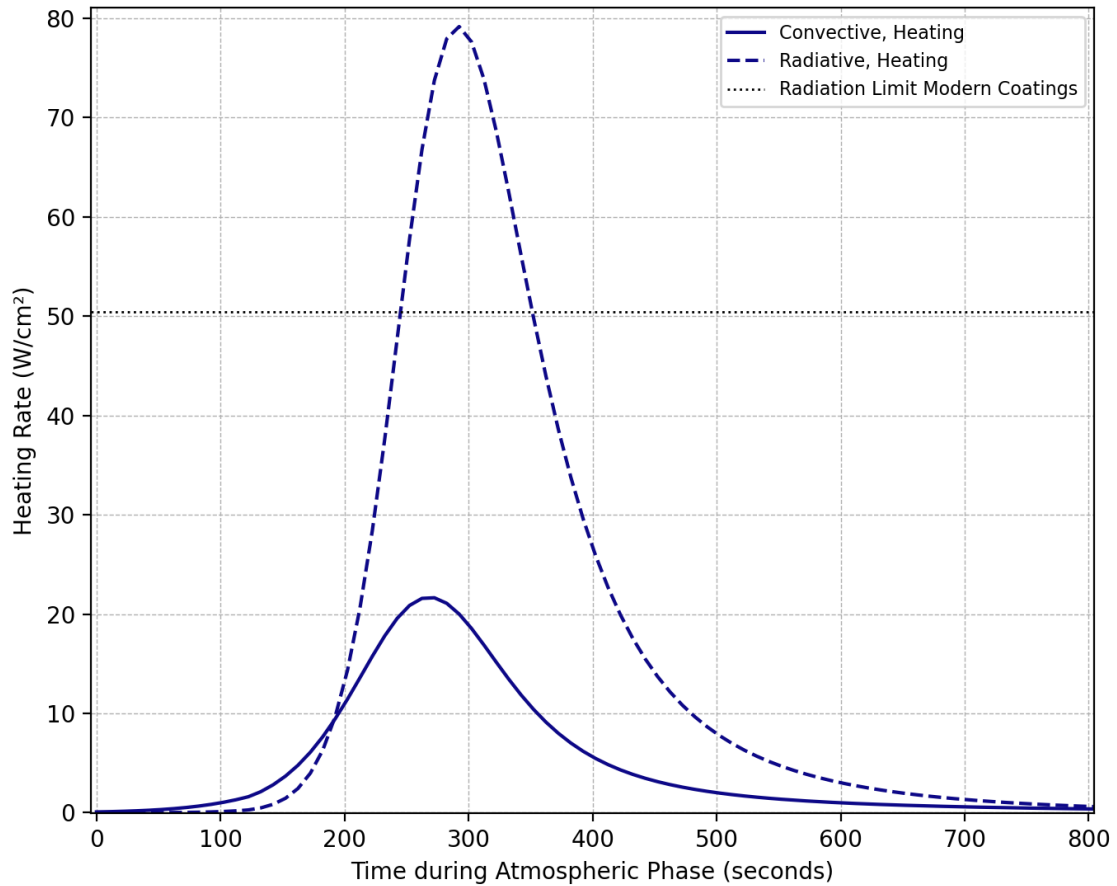


Figure 4.11: Heatflux during aerocapture program model

5 Conclusion

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A Extra Figures

A.1 View of all Saturn Moons

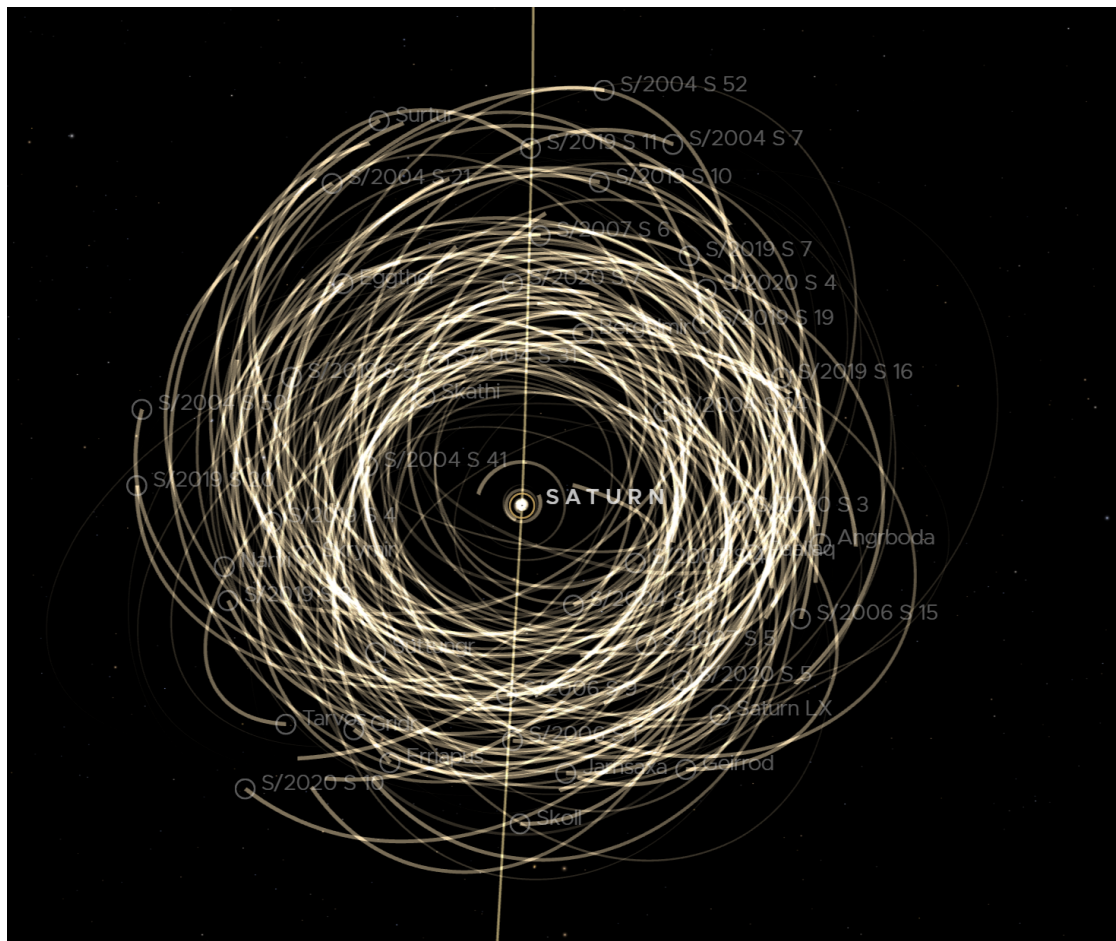


Figure A.1: Saturn and all known moons (Barnett, 2025)

A.2 Complete Trajectory Plot

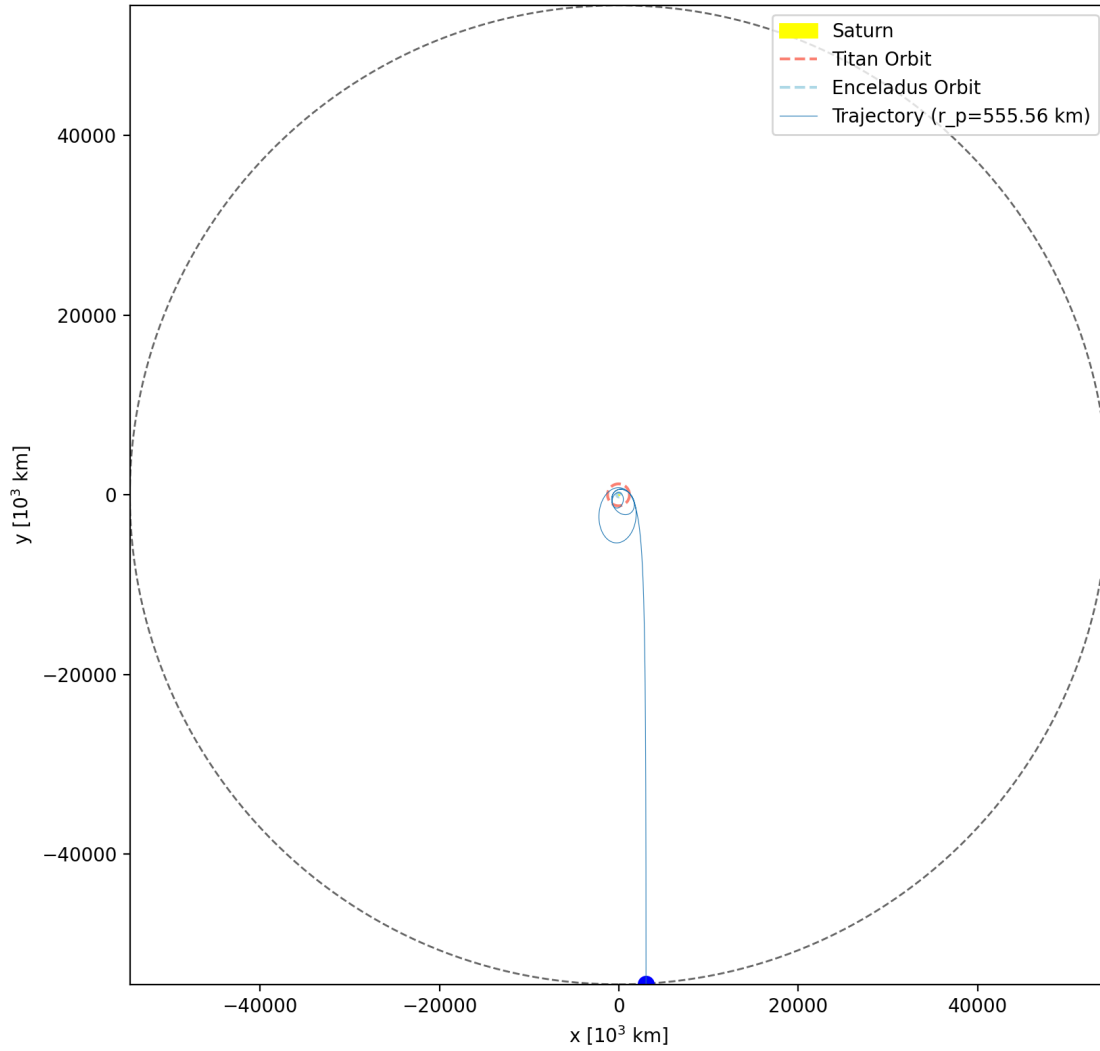


Figure A.2: Complete Trajectory Plot (own work)

A.3 Deviation in Aerothermodynamics

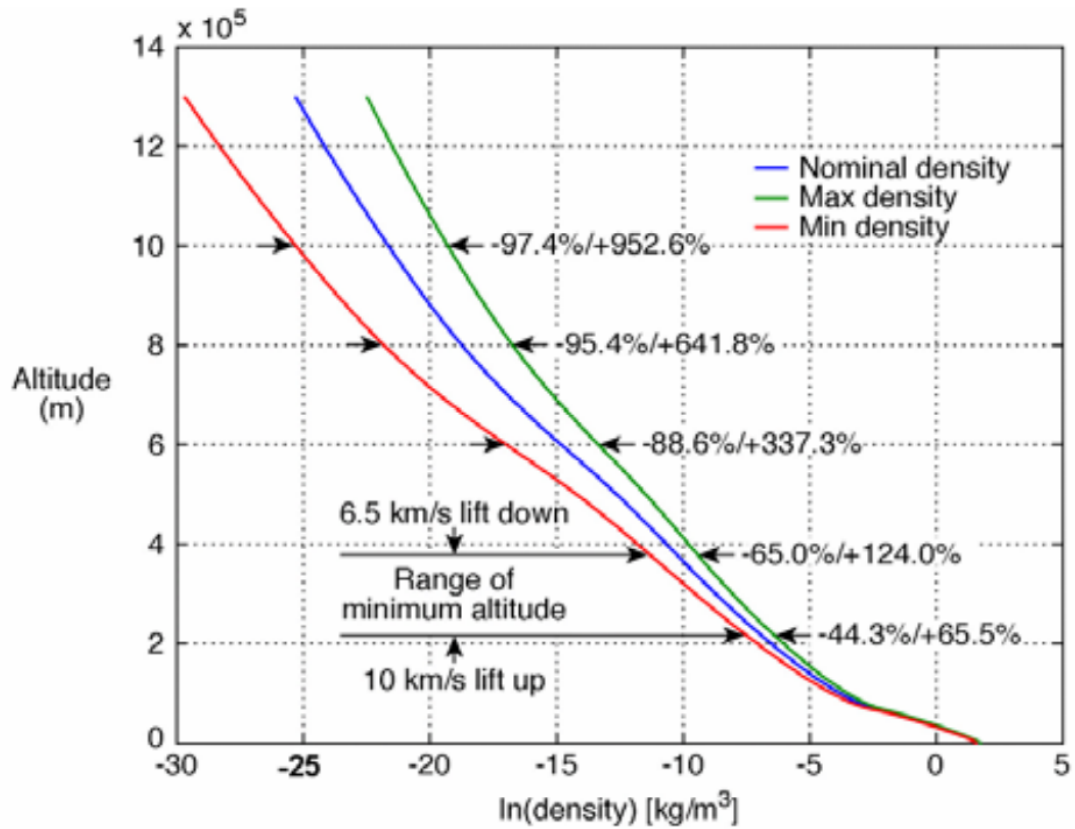


Figure A.3: Atmosphere model NASA aerothermodynamics study (Laub, 2012)

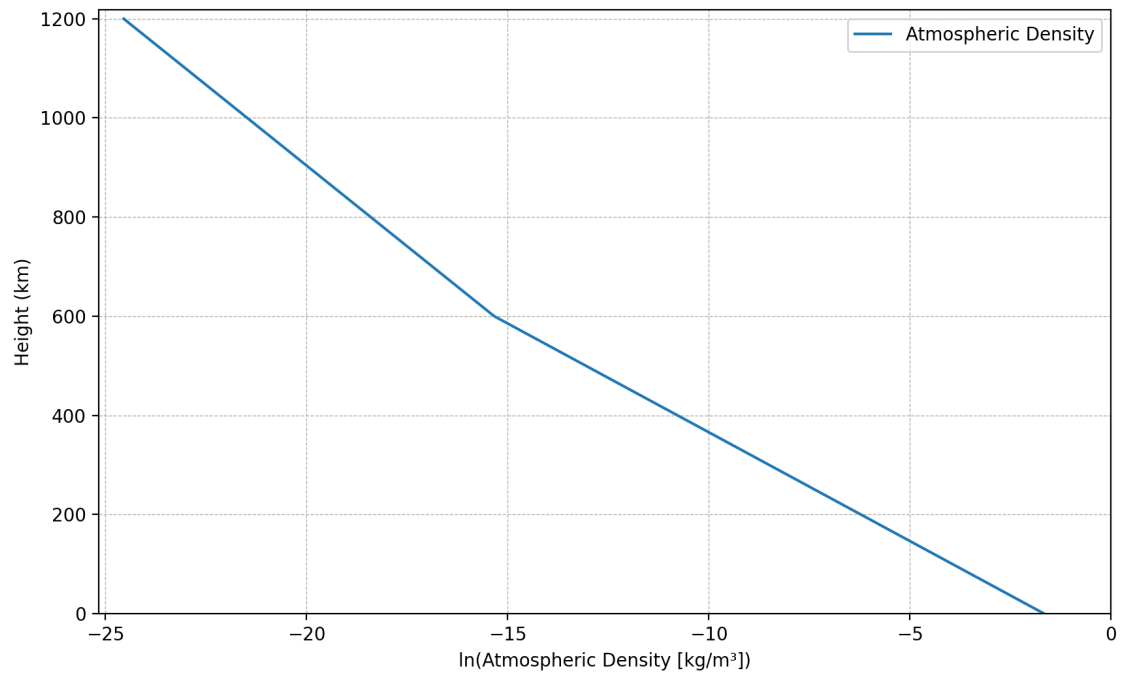


Figure A.4: Atmosphere model simulation program (own work)

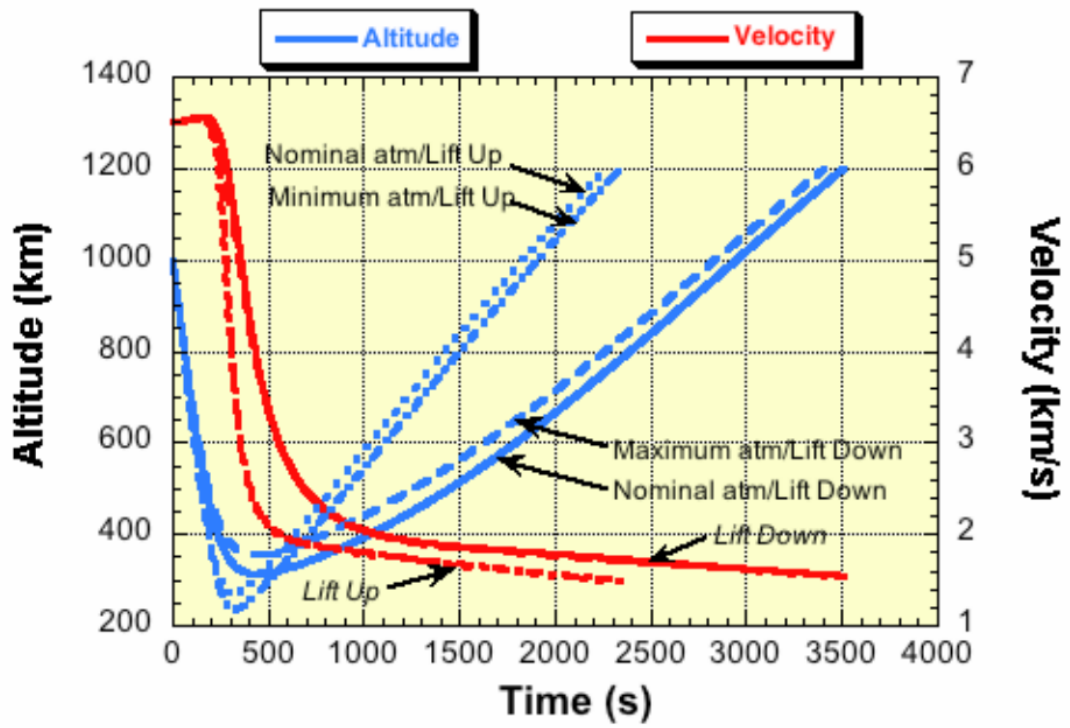


Figure A.5: Altitude & velocity during aerocapture NASA aerothermodynamics study (Laub, 2012))

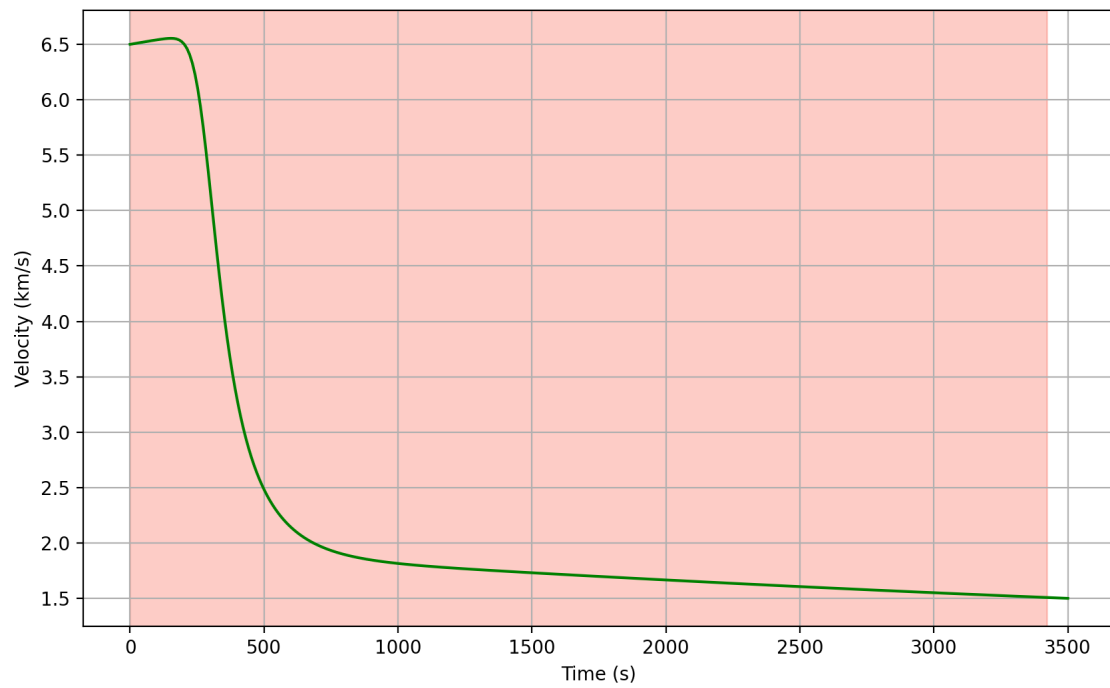


Figure A.6: Velocity during aerocapture simulation program (own work)

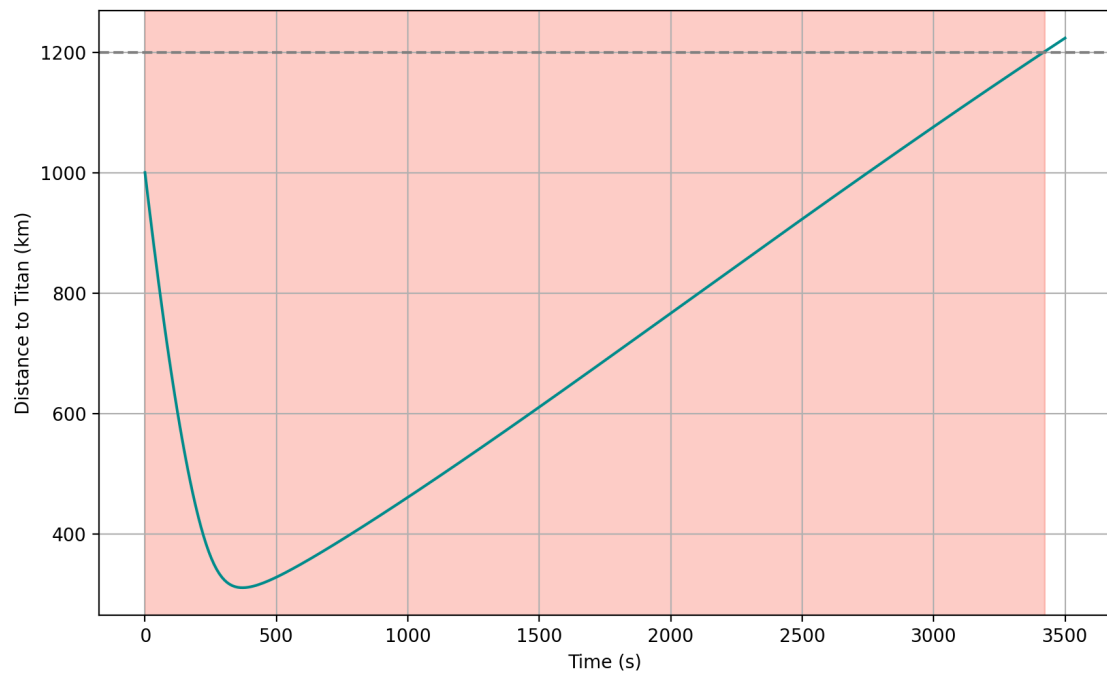


Figure A.7: Altitude during aerocapture simulation program (own work)

B Reference Frames

B.1 Inertial Cartesian Reference Frame

B.2 Orbit Reference Frame

C Python Code

The entire python code is available for download at: <https://github.com/CedricKalscheuer/Aerobraking>.

C.1 Launch Delay Iteration Process

C.2 Adjust Aerobrake Altitude Process